

1. GENETIC DISEASE ANALYSIS



**Trabajo realizado por:
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 U.S. Department of Health & Human Services

 National Institutes of Health

 National Center for Advancing Translational Sciences

GARD Genetic and Rare Diseases Information Center

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Treacher-Collins syndrome

Other Names: Franceschetti syndrome; Franceschetti-Klein syndrome; mandibulofacial dysostosis without limb anomaly syndrome

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1. What is the disease?



Treacher-Collins syndrome | About the Disease | GARD

Find symptoms and other information about Treacher-Collins syndrome.

 info.nih.gov

What Causes This Disease?

[Genetic Mutations](#)

Known Genetic Mutations

Known Genetic Mutations

What Is A Genetic Mutation?

 Typical Gene

 Genetic Mutation

Treacher-Collins syndrome is caused by genetic mutations in the following known gene(s):

POLR1D, TCOF1, POLR1B, POLR1C

[Learn more about these genes by searching them on the National Library of Medicine \(NLM\).](#)

Given these known genetic mutation(s), you may want to ask your health care team if genetic testing is right for you. Genetic tests are laboratory tests that use samples of blood, saliva, or other tissues to help identify changes in genes, chromosomes, or proteins. Genetic testing can help confirm or rule out a suspected genetic disease or can provide other useful information to your health care team. [Learn more about genetic testing from the National Human Genome Research Institute \(NHGRI\).](#)

2. How is the genetics influencing in the disease? Is it because a mutation, loss of function or any other reason?



POLR1C gene: MedlinePlus Genetics

The POLR1C gene provides instructions for making one part (subunit) of two related enzymes called RNA polymerase I and RNA polymerase III. Learn about this gene and related health conditions.



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POLR1C gene

RNA polymerase I and III subunit C

Normal Function

▼

Health Conditions Related to Genetic Changes

^

Treacher Collins syndrome

^

At least six mutations in the *POLR1C* gene have been identified in people with Treacher Collins syndrome, a condition that affects the development of bones and other tissues of the face. These mutations appear to alter the structure and function of the POLR1C protein, which reduces the amount of functional RNA polymerase I and RNA polymerase III in cells. Consequently, less rRNA is produced. Researchers speculate that a shortage of rRNA may trigger the self-destruction (apoptosis) of certain cells involved in the early development of facial bones and tissues. The abnormal cell death could underlie the specific problems with facial development found in Treacher Collins syndrome. However, it is unclear why the effects of a reduction in rRNA are limited to facial development.

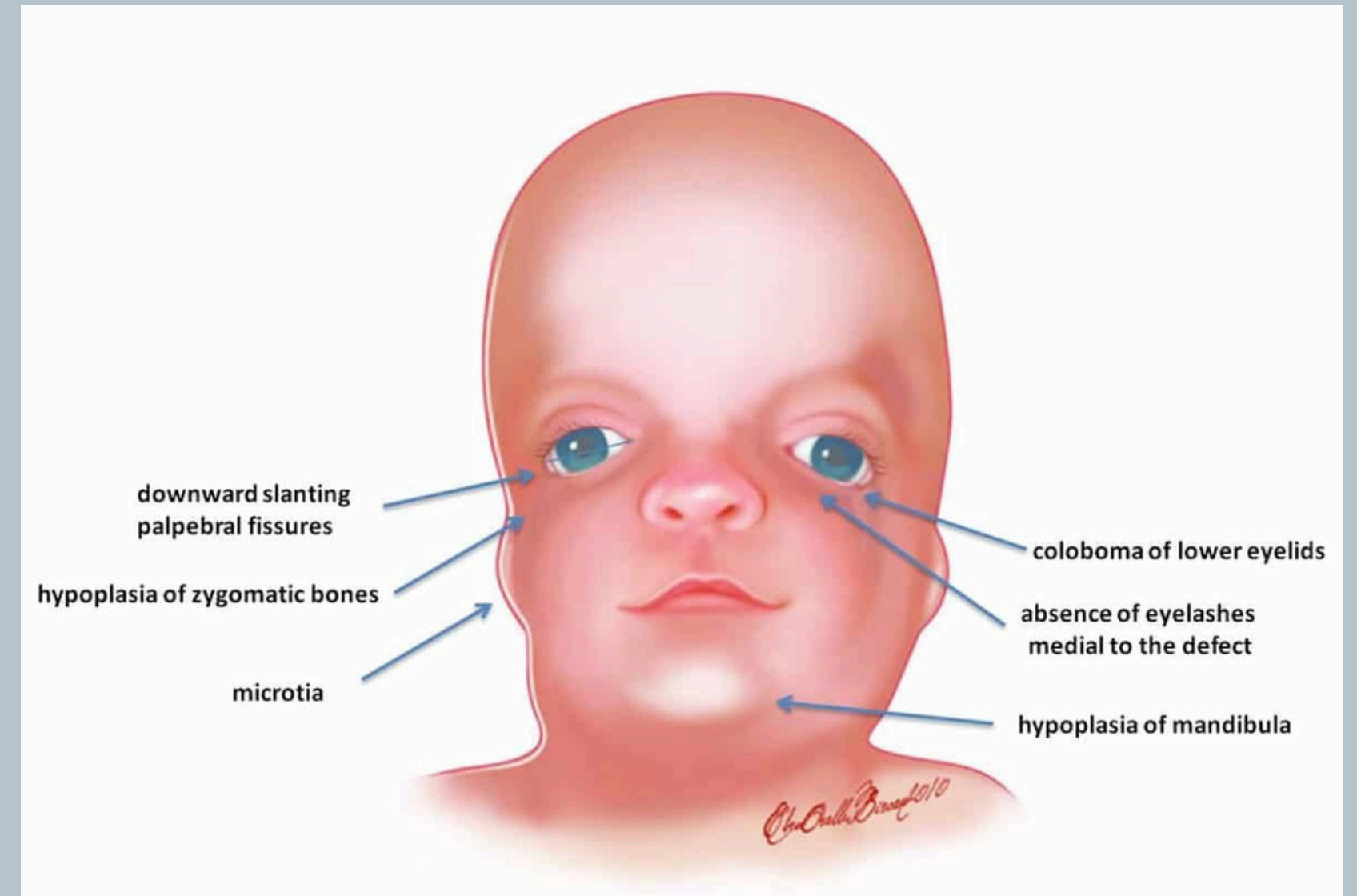
Treacher Collins Syndrome

The disease chosen was Treacher Collins Syndrome, a disease characterized by the development of bones and tissues in the skull, leading to cranioencephalic abnormalities, like ear abnormalities, down slanting eyes or jaw deformities, like micrognathia, etc.

The pathology is due to **ribosomal abnormalities**, which can cause ribosome biogenesis deficiency, neural crest cell apoptosis and oxidative stress and DNA damage.

Several genes can cause this pathology, such as TCOF1, which encodes Treacle, a protein involved in the biogenesis of the ribosomes or POLR1C and POLR1D, which encode for certain subunits of RNA polymerase I and III, responsible for rRNA and some tRNA synthesis.

I'm going to focus on **POLR1C**. Mutations in this gene alter the structure and function of its corresponding protein, reducing the quantity of RNA polymerase being created, and as consequence, the corresponding rRNA.



3. Are there any clinical trials currently being conducted on this disease? Pick one, explain what are they doing, current status and the country

clinicaltrials.gov/study/NCT04931056?cond=Treacher%20Collins%20Syndrome&rank=1

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

PRS Login

ClinicalTrials.gov

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Completed ⓘ

A Post Market Clinical Follow-up Study on Biomet Microfixation HTR PEKK (Midface), Facial & Mandibular Plates.

ClinicalTrials.gov ID ⓘ NCT04931056

Sponsor ⓘ Zimmer Biomet

Information provided by ⓘ Zimmer Biomet (Responsible Party)

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Noack Watt KE, Achilleos A, Neben CL, Merrill AE, Trainor PA (2016) The Roles of RNA Polymerase I and III Subunits Polr1c and Polr1d in Craniofacial Development and in Zebrafish Models of Treacher Collins Syndrome. PLOS Genetics 12(7): e1006187. <https://doi.org/10.1371/journal.pgen.1006187>



RESEARCH ARTICLE

The Roles of RNA Polymerase I and III Subunits Polr1c and Polr1d in Craniofacial Development and in Zebrafish Models of Treacher Collins Syndrome

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The Roles of RNA Polymerase I and III Subunits Polr1c and Polr1d in Craniofa  **Search**

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Affiliations  expand

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According to clinicaltrials.gov, there are no actual studies in Treachers Collins Syndrome underway regarding the POLR1C gene. However, by searching for past studies in PubMed, I was able to find one that probed to be very interesting.

This research was done in several **institutes of USA**, and it focuses on the **roles of POLR1C and POLR1D**. Using mutated zebrafish, which did not express these genes in the embryonic and developmental stage when they are needed, they were able to see the differential development of the craniofacial region. With the information obtained from the study, they correlated the results to Treachers Collins syndrome, where these genes are also affected.

They studied gene expression analysis by ISH, cartilage formation by staining, cell death by TUNEL assays, ribosome biogenesis by polysome profiling, and the role of Tp53 in cell death in this situation by genetic manipulation.

The main results of the study were that both POLR1C and POLR1D were **highly expressed in craniofacial tissues during developments**, and in those **mutants with loss of function, cranioskeletal abnormalities** similar with the ones in Treachers Collins syndrome were exhibited. This was due to the disruption on ribosome biogenesis, result of the mutation, that led to a deficient migration of the neural crest cells and a **TP53-dependent neuroepithelial cell death**.

As they saw that Tp53 was the primary participant which affected in the cell's death, they studied how the inhibition of this gene could affect the pathology. They saw that, once blocked TP53, an improvement could be seen in the craniofacial development, suggesting that this inhibition could be a potentially therapeutic approach.

1. Obtain nucleotide and protein sequence

<https://www.ncbi.nlm.nih.gov/gene/9533>

GENE

Was this help

POLR1C – RNA polymerase I and III subunit C

Homo sapiens (human)

Also known as: AC40, HLD11, RPA39, RPA40, RPA5, RPAC1, RPC40, TCS3

Gene ID: 9533

RefSeq products

Orthologs

Genome Data Viewer

Nucleotide sequence

```
1 >ref|NM_203290.4|:1-1275 Homo sapiens RNA polymerase I and III subunit C (POLR1C), transcript variant 1, mRNA
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3 GGGAGTTTGGGGTTCGCAATGTCCATACTACTGACTTTCCCGGTAACATTCCGGTTATGATGATGCCTG
4 GGACCAGGACCGCTTCGAGAAGAATTTCCGTGTGGATGTAGTACACATGGATGAAAACACTGAGTTT
5 GACATGGTGGGAATTGACGCAGCCATTGCCAATGCTTTTCGACGAATTCTGCTAGCTGAGGTGCCAACTA
6 TGGCTGTGGAGAAGGTCCTGGTGTACAATAATACATCCATTGTTTCAGGATGAGATTCTTGCTCACCGTCT
7 GGGGCTCATTCCCATTGCTGATCCCCGTCTTTTTCAGTATCGGAACCAAGGAGATGAAGAAGGCACA
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12 CAGTGGCAACAGCCAGTTACAGGCTCCTGCCAGACATCACCTGCTTGAGCCCGTGGAAGGGGAGGCAGC
13 TGAGGAGTTGAGCAGGTGCTTCTCACCTGGTGTATTGAGGTGCAGGAAGTCCAAGGTAAAAAGGTGGCC
14 AGAGTTGCCAACCCCGGGCTGGATACCTTCAGCAGAGAAATCTTCCGGAATGAGAAGCTAAAGAAGGTTG
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16 GCTGGTGAGTGAAGCCATCAAAGTACTGATGGGGAAGTGCCGGCGCTTCTTGATGAAGTAGATGCGGTT
17 CAGATGGACTGAGCTTGGATGCTTCTGAGGCAAGCTGAAGCTTTGGGTTCTGACTGACCCACCCTACAGG
18 ACTGCTGAACAGAGAGCCAGTGTGACTAGGGATCCTGAGTTTCTGGGACAATTCCAGCTTTAATCAAT
19 ACATTTTGTTAAATGTGCCATAAAATGAGACTTTTACGCCTTTATAAGGCCTTAGATGTAAATAAACTC
20 ACCCAAACAAAAAA
21
```

Protein sequence

```
NP_humano.fa
1 >ref|NP_976035.1|:1-346 DNA-directed RNA polymerases I and III subunit RPAC1 isoform 1 [Homo sapiens]
2 MAASQAVEEMRSRVVLGEFGVRNVHTTDFPGNYSYDDAWDQDRFEKNFRVDVHMDENSLEFDMVGIDA
3 AIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPRLFEYRNQGDEEGTEIDTLQF
4 RLQVRCTRNPAAKDSDPNELYVNHKVYTRHMTWIPLGNQADLFPEGTIRPVHDDILIAQLRPGQEIDL
5 LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGEAAEELSRCSFPGVIEVQEVQGKKVARVANPRL
6 DTFSREIFRNEKLKKVRLARVRDHYIFSVESTGVLPPDVLVSEAIKVLGMGKCRFLDELDAVQMD
7
```


2. From the nucleotide sequence: obtain the three main orthologues for this gene in plants, bacteria and mammals (excluding Homo sapiens) -> for each orthologue, obtain the nucleotide sequence.



No orthologues in bacteria

No orthologues in plants

Your results are filtered to match records that include: bacteria (taxid:2)

Job Title	Nucleotide Sequence
RID	X3H9MDH6016 Search expires on 03-14 01:32 am Download All ▼
Program	? Citation ▼
Database	nt See details ▼
Query ID	lcl Query_815069
Description	None
Molecule type	dna
Query Length	1275
Other reports	?

No significant similarity found. For reasons why,[click here](#)

<https://blast.ncbi.nlm.nih.gov/Blast.cgi#>

Your results are filtered to match records that include: plants (taxid:3193)

Job Title	Nucleotide Sequence
RID	X3H9MDH6016 Search expires on 03-14 01:32 am Download All ▼
Program	? Citation ▼
Database	nt See details ▼
Query ID	lcl Query_815069
Description	None
Molecule type	dna
Query Length	1275
Other reports	?

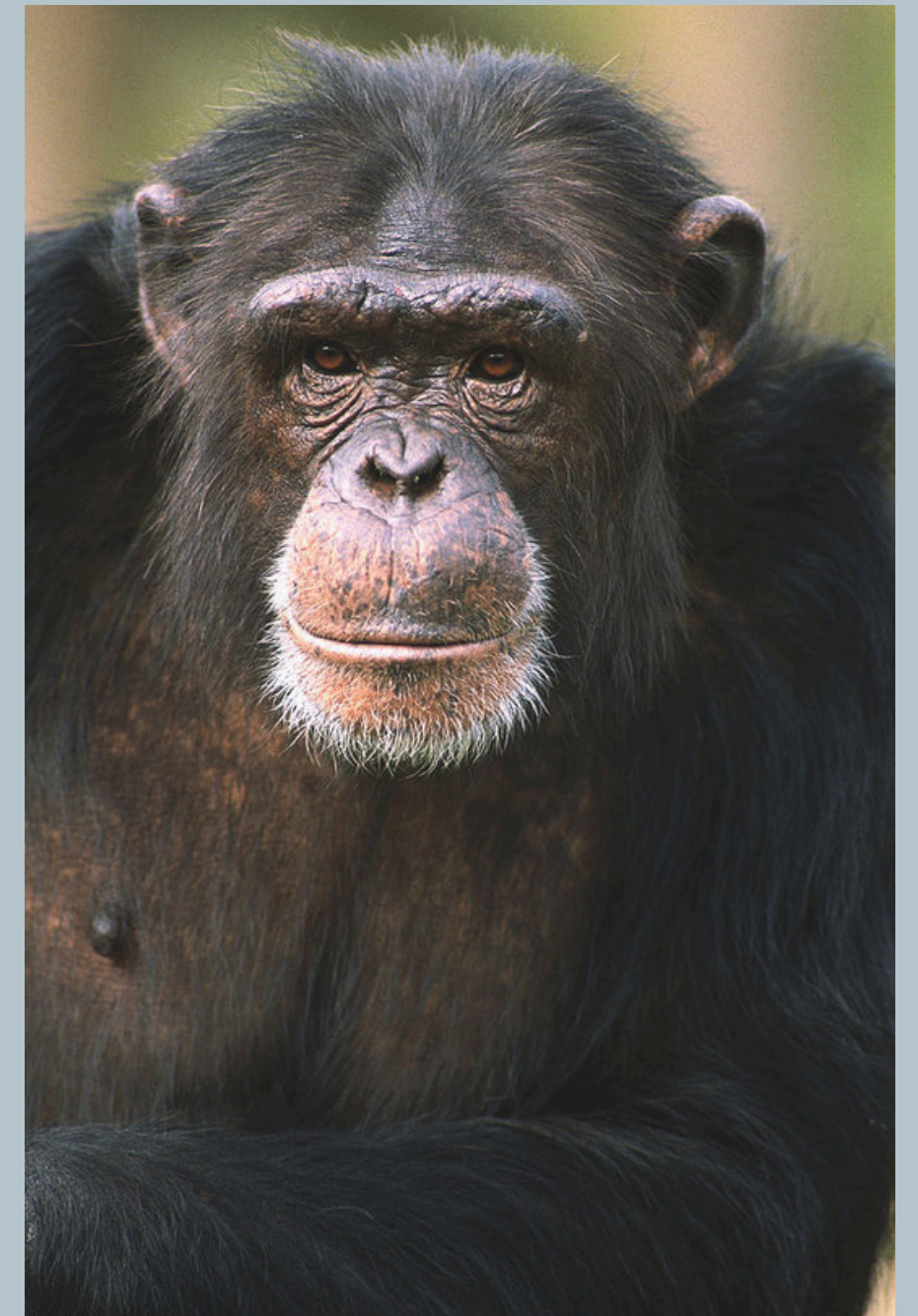
No significant similarity found. For reasons why,[click here](#)

Orthologues

As I was not able to obtain orthologues in bacteria or plants, I used different taxids.

Pan troglodytes

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seqdump (5).txt seqdump (6).txt seqdump (7).txt seqdump (8).txt seqdump (9).txt seqdump (10).txt seqdump (11).txt seqdump (12).txt
1 >XM_001136605.7 PREDICTED: Pan troglodytes RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
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3 AGGCGGCGCGTTGCCACGGAGACGGAGCAGGACTACTAGGACGCACGCGGAGATAGAACCTCTAGTCTCGTGGAGAGAT
4 TGAAGATGGCGGCTTCTCAGGCGGTGGAGGAAATGCGGAGCCGCGTGGTTCTGGGGGAGTTTGGAGTTCGCAATGTCCAT
5 ACTACTGACTTTCCCGGTAACATATCCGGTTATGATGATGCCTGGGACCAGGACCCTTCGAGAAGAATTTCCGTGTGGA
6 TGTAGTACACATGGATGAAAACACTGAGGTTTGACATGGTGGGAATTGACGCAGCCATTGCCAATGCTTTTCGACGAA
7 TTCTGCTAGCTGAGGTGCCAACTATGGCTGTGGAGAAGGTCCTGGTGTACAATAATACATCCATTGTTTCAGGATGAGATT
8 CTTGCTCACCGTCTGGGGCTCATTCCCATTCATGCAGATCCCCGTCTTTTGTAGTATCGGAACCAAGGAGATGAAGAAGG
9 CACAGAGATAGATACTCTACAGTTTCGTCTCCAAGTCAGATGCACTCGGAACCCCATGCTGCTAAAGATTCTCTGACC
10 CCAACGAACTGTACGTGAACCACAAAGTGTATACCAGGCATATGACATGGATCCCCCTGGGGAACCAGGCTGATCTCTTT
11 CCAGAGGGCACTATCCGACCAGTGCACGATGATATCCTCATCGCTCAGCTGCGGCCTGGCCAAGAAATTGACCTGCTCAT
12 GCACTGTGTCAAGGGCATTGGCAAAGATCATGCCAAGTTTTACCAGTGGCAACAGCCAGTTACAGGCTCCTGCCAGACA
13 TCACCCTGCTTGAGCCCGTGGAAGGGGAGGCAGCTGAGGAGTTGAGCAGGTGCTTCTCACCTGGTGTTATTGAGGTGCAG
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16 ATGTGCTGGTGAGTGAAGCCATCAAAGTACTGATGGGGAAGTGCCGGCGCTTCTTGGATGAACTAGATGCGGTTTCAGATG
17 GACTGAGCTTGGATGCTTCTGAGGCAAGCTGAAGCTTTGGGTTCTGACTGACCCACCTACAGGACTGCTGAACAGAGCC
18 CAGTGTGACTAGGGATCCTGAGTTTTCTGGGACAATTCCAGCTTTAATCAATACATTTTGTAAATGTGCCATAAAATGA
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20
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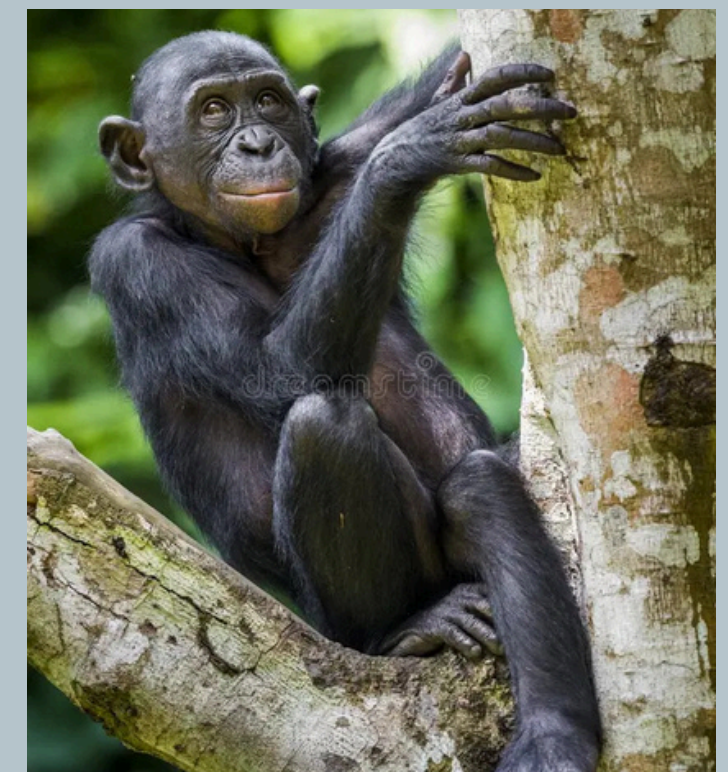

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seqdump (5).txt seqdump (6).txt seqdump (7).txt seqdump (8).txt seqdump (9).txt seqdump (10).txt seqdump (11).txt
C:\Users\juanc\Downloads\NM_203290.4.fasta PREDICTED: Pongo abelii RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
2 GTGGACGGGGCGGGGCAAAGTCTCGCGATATTTAAGATTCCAGGAGGCGGTGCGTTGCCACGGAGACGGAGCAGGGGCTA
3 CTAGGACGCACGCGTCAGATAGAGCCTCTAGTCTTGTGGAGAGATTGAAGATGGCGGCTTCTCAGGCGGTGGAGGAAATG
4 CGGAGTCGCGTGGTTCTAGGGGAGTTTGGGGTTCGCAATGTTCAATCACTGACTTTCCCGGTAACCTATTCGGTTATGA
5 TGATGCCTGGGACCAGGACCGCTTCGAGAAGAATTTCCGTGTGGATGTAGTACACATGGATGAAACTCACTGGAGTTTG
6 ACATGGTGGGAATTGACGCAGCCATTGCCAATGCTTTTCGACGAATTCTGCTAGCTGAGGTGCCAACTATGGCTGTGGAG
7 AAGGTCCTGGTGTACAATAATACATCTATTGTTTCAGGATGAGATCCTTGCTCACCGTCTGGGGCTCATTCCCATTTCATGC
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10 AGGCATATGACATGGATCCCCCTGGGGAACCAAGGCTGATCTCTTCCAGAGGGCACTATCCGACCAGTGCACAATGACAT
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13 GAGGAGTTGAGCAGGTGCTTCTCACCTGGTGTATTGAGGTGCAGGAAGTCCAAGGTAAAAAGGTGGCCAGAGTTGCCAA
14 CCCCCGGCTGGATACCTTCAGCAGAGAAATCTTCCGGAATGAGAAGCTAAAGAAGGTTGTGAGGCTTGCCCGGGTTCGAG
15 ATCATTATATCTTCTCTGTTGAGTCAACAGGGGTGTTGCCACCAGATGTGCTGGTGAAGCCATCAAAGTACTGATG
16 GGAAGTGCCGGCGCTTCTTGATGAAGTAGATGCGGTTGAGTGGAGTGAAGCCATCAAAGTACTGATG
17 TTTGGGTTCTGACTGACCCACCCTACAGGACTGCTGTACAGAGAGCCCAGTGTGACTAGGGATCCTGAGTTTTCTGGGAC
18 AATTCCAGCTTTAATCAATATATTTTGTAAATGTGCCATAAAATGAGACTTTTTTAGGCCTTTATAAGGCCTTAGATGTA
19 AATAAACTACCCCAAACAAAAACA
20
```

Pongo abelii



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1 >XM_034962155.2 PREDICTED: Pan paniscus RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
2 CGGCGTTGTGCGGAAGGCCGCGCAGATGTCCCGCACCTCGGCGCCGGACAGCGGCTCCCCGAAATCTGGATGGAGCTG
3 GGCAGCATCGTGCCACGGCTGGGCCTCCGGGTACGGGGCCGGAACGGAGATTCCAGGAGGCGGCGGCTTGCCACGGAGA
4 CGGAGCAGGACTACTAGGACGCACGCGGAGATAGAACCTCTAGTCTCGTGGAGAGATTGAAGATGGCGGCTTCTCAGGC
5 GGTGGAGGAAATGCGGAGCCGCGTGGTTCTGGGGGAGTTTGGAGTTCGCAATGTCCATACTACTGACTTTCCCGGTAAC
6 ATTCGGTTATGATGATGCCTGGGACCAGGACCGCTTCGAGAAGAATTTCCGTGTGGATGTAGTACACATGGATGAAAC
7 TCACTGGAGTTTGACATGGTGGGAATTGACGCAGCCATTGCCAATGCTTTTCGACGAATTCTGCTAGCTGAGGTGCCAAC
8 TATGGCTGTGGAGAAGGTCCTGGTGTACAATAATACATCCATTGTTTCAGGATGAGATTCTTGCTCACCGTCTGGGGCTCA
9 TTCCCATTTCATGCTGATCCCCGTCTTTTGTAGTATCGGAACCAAGGAGATGAAGAAGGCACAGAGATAGATACTCTACAG
10 TTTCTGCTCCAAGTCAGATGCACTCGGAACCCCATGCTGCTAAAGATTCTCTGACCCCAACGAACTGTACGTGAACCA
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12 TGCACGATGATATCCTCATCGCTCAGCTGCGGCCTGGCCAAGAAATTGACCTGCTCATGCACTGTGTCAAGGGCATTGGC
13 AAAGATCATGCCAAGTTTTACCAGTGGCAACAGCCAGTTACAGGCTCCTGCCAGACATCACCTGCTTGAGCCCGTGGA
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16 GCCCGGGTTCGAGATCATTATATCTTCTCTGTTGAGTCAACGGGGGTGTTGCCACCAGATGTGCTGGTGAAGCCAT
17 CAAAGTACTGATGGGGAAGTGCCGGCGCTTCTTGATGAAGTAGATGCGGTTGAGTGGAGTGAAGCCAT
18 GGCAAGCTGAAGCTTTGGGTTCTGACTGACCCACCCTACAGGACTGCTGAACAGAGTCCAGTGTGACTAAGGATCCTGAG
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20 CCTTAGATGTAAATAAACTACCCCAAACAAAAAAA
21
```

Pan paniscus




```
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3 GCCCAGGCTGTGGAGGAAATGCGGAGCCGCGTGGTTCTAGGGGAGTTCGGGATTTCGCAATGTTTCATACCACCGACTTTCC
4 CGGTAAC TATTTCGGGCTACGATGATGCTTGGGACCAGGACCGCTTCGAGAAGAATTTCCGTGTGGATGTGGTACACATGG
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6 GTGCCAACCATGGCTGTGGAGAAGGTCCTGGTGTACAACAACACTTCCATTGTCCAGGATGAGATCCTTGCTCACC GCTT
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19
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Hippopotamus amphibius kiboko



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4 GAAGAATTTCCGTGTGGATGTGGTACACATGGATGAAAAC TCA TTGGAGTTTGACATGGTGGGAATCGACGCAGCCATAG
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12 GGTGTTATTGAGCTGCAGGAAGTCCAAGGTAAAAAGGTGGCCAGAGTTGCCAATCCCCGGCTCGATACCTTTAGCAGGGA
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22 TAAATTCACATTCAAAAAGCTTTTT
23
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Diceros bicornis




```
seqdump (5).txt seqdump (6).txt seqdump (7).txt seqdump (9).txt seqdump (10).txt seqdump (11).txt seqdump (12).txt seqdump
1 >XM_046639285.1 PREDICTED: Equus quagga RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
2 GGGACGTGGACGGAGGCGGAGCGGAGTCTCGCGATATTTAAGATTCCAGGCGGTGGTGCCTTGCCACGGAGACGGAGGAG
3 GGCGACGAGAACGCACGCGCCGGTTTGTATGCTCTAGTCACGTGCAGGCGTTGAAGATGGCGGCGGCCAGGCGGTGGAAG
4 AAATGCGGAGCCCGTGGTTCTAGGGGAGTTTCGGGGTTCGCAATGTTTCATACGACGGACTTTCCCGGCAACTATTCCGGC
5 TATGATGATGCCTGGGACCAGGACCACTTCGAGAAGAATTTCCGTGTGGATGTAGTACACATGGATGAAAACCTCATTGGA
6 ATTTGACATGGTGGGAATCGACGCGGCCATTGCCAATGCTTTTCGACGCATTCTATTAGCTGAGGTGCCAACCATGGCTG
7 TGGAAAAGGTCCTGGTGTACAACAACACATCCATTGTCCAGGATGAGATCCTTGCTCATCGCTTGGGGCTCATTCCCATT
8 CACGCTGATCCCCGTCTTTTTGAATATCGGAACCAAGGAGATGAAGAAGGCACAGAGATAGATACTCTGCAGTTCCGGCT
9 GCAGGTTAGGTGTACTCGGAACCCCCATGCTGCTAAAGATTCTCTGACCCCACTGAAGTCTACGTGAACCACAAAGTGT
10 ACACCAGACACATGATATGGGTGCCCTTGGGGAACCAGGCTGACCTTTTCCAGAGGGCACTATCCGACCAGTGCATGAT
11 GATATCCTCATTGCTCAGCTGCGGCCTGGCCAGGAGATTGACCTGCTCATGCACTGTGTCAAGGGCATTGGCAAGGATCA
12 TGCCAAGTTTTTCACCTGTGGCCACAGCCAGCTATAGGCTCCTGCCAGACATCACCCTGCTTGAGCCTGTGGAAGGGGATG
13 CGGCTGAGGAGCTGAGCCGGTGCTTCTCGCCGGGTGTTATTGAGGTACAGGAAGTCCAAGGTAAAAAGGTGGCCAGAGTT
14 GCCAATCCCCGGCTAGATACCTTTAGCAGGGAAGTCTTCCGGCATGAGAAGCTAAAGAAGGTTGTGCGGCTGGCTCGGGT
15 TCGGGACCATTATATCTTCTCTGTGGAGTCAACAGGGGTGTTGCCACCAGATGTGCTGGTGAGTGAAGCCATCAAGGTAC
16 TGATGGGGAAGTGCCGGCGGTTCTTGGATGAGCTAGAGGCGGTTTCCAGATGGACTGAAGTTTCCAGACTTACTGGCATGGC
17 TGGCCTCGGATGCTTGCTCCTGAGGCAAATGGAAGATTTGGGATTCTGGCCTGCCCCACAGGAGTGCTGTGGAGAGCCC
18 AGTGTAACCAGGGGTCTTCGGTTTTCTGGGACAATCACCATTCTGAGTCAATACATTTTGTTAAATGTGCCATAA
19 AATGTGACTTTTATGCCTTTGTAAGGCCTTAATGTAAATAAATTCACCCTCAACAAAAAACCTTTTTTTTTT
20
```

Equus quagga



```
seqdump (8).txt NM_203290.4.fa seqdump (5).txt seqdump (6).txt seqdump (7).txt seqdump (9).txt seqdump (10).txt seqdump (11).tx
1 >XM_040473614.1 PREDICTED: Puma yagouaroundi RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
2 CCACGGAGACGGAGGTGGGGGGCGCGCGGCCAGTTTGCTGCCCTAGTCACGTGGAGGGGTGAAGATGGCGGCTGCTCA
3 GGCAGGTGGAAGAAATGCGGACCCGCGTGTTCTAGGAGAGTTTCGGGGTTCGCAATGTTTCATACCACTGACTTTCCTGGTA
4 ACTATTCGGGTATGATGATGCCTGGGACCAAGACCGCTTCGAGAAGAATTTCCGTGTGGATGTAGTACACTTGGATGAA
5 AACTCATTGGAGTTTGACATGGTGGGAATTGATGCAGCCATTGCCAATGCTTTTCGGCGCATTTTATTAGCTGAGGTGCC
6 AACCATGGCTGTGGAAGGTCCTGGTGTACAACAACACGTCCATTGTCCAGGATGAGATCCTTGCTCATCGCTTGGGGC
7 TCATTCCCATTTCATGCTGATCCCCGTCTTTTTGAATATCGGAACCAAGGAGATGAAGAAGGTACAGAGATAGATACTCTA
8 CAGTTTCGGCTTCATGTCAGGTGCACTCGGAACCCCCATGCTGCTAAAGATTCTCTGACCCCAATGAACTCTATGTGAA
9 CCATAAAGTGACACCAGGCACATGACATGGGTGCCCCCTGGGGAATCAGGCTGACCTCTTCCCAGAGGGCACGATCAGAC
10 CAGTGCATGATGATATTCTCATCGCTCAGCTACGGCCTGGCCAGGAGATTGACTTACTCATGCACTGTGTCAAGGGCATT
11 GGCAAGGATCATGCCAAGTTTTACCTGTGGCAACAGCCAGTTACAGGCTTCTGCCAGACATCACCCTGCTTGAGCCTGT
12 GGAAGGGGATGCAGCGGAGGAGCTGAGGCGGTGCTTCTCGCCTGGTGTATTGAGGTGCAGGAAGTCCAAGGTAAAAAGG
13 TGGCCAGAGTTGCCAATCCCCGGCTAGATACCTTCAGCAGAGAAGTGTTCCGGAACGAGAAGCTAAAGAAGGTTGTACGA
14 CTTGCTCGAGTTCGGGACCACTATATCTTTTCTGTGAGTCAACAGGGGTGTTGCCGCCAGAGGTGCTAGTGAGTGAAGC
15 CATCAAGGTAAGTGGGGAAGTGCCAACGGTCTTGGATGAACTAGATGCAGTTCAGATGGACTGAGGTTTGCATGGGA
16 CTTACCGGTATGGCTGTGCTTGGATACTCAAGTATCCTAGCTGAAGATTTTGGGTCCTGACTTGCTGGTCACAGGACTGC
17 TAGAGTCCAGTGTGACTGCAGGTCCTGGGTTTTCTGGGACAATCCAGCCTTATTTTCAGTCAATACATTTTGTTAAATG
18 TGCCACAAAATGTGACTTCCTGCCTTTGTAAGGCCTTAATATAAATAAACTCACATCCAAATAAAAAACCTTTTTGCCAG
19 AGACTA
20
```

Puma yagouroundi




```
seqdump (8).txt  NM_203290.4.fa  seqdump (5).txt  seqdump (6).txt  seqdump (7).txt  seqdump (9).txt  seqdump (10).txt  seqdun
1  >XM_015071216.3 PREDICTED: Acinonyx jubatus RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
2  ATCCCGGGTGCGTTGCCACGGAGACGGAGGTGGGGGGCGCGCGCCAGTTTGTCTGCCCTAGTCACGTGGAGGGGTTGAA
3  GATGGCGGGCTGCTCAGGCGGTGGAAGAAATGCGGACCCGCGTGGTTCTAGGAGAGTTCGGGGTTCGCAATGTTTCATACCA
4  CTGACTTTCTCTGGTAACTATTCGGGTATGATGATGCCTGGGACCAAGACCGCTTCGAGAAGAATTTCCGTGTGGATGTA
5  GTGCACTTGGATGAAAACCTCATTGGAGTTTGACATGGTGGGAATTGATGCAGCCATTGCCAATGCTTTTCGGCGCATTTT
6  ATTAGCTGAGGTGCCAACCATGGCTGTGGAAAAGGTCTTGGTGTACAACAACACGTCCATTGTCCAGGATGAGATCCTTG
7  CTCATCGCTTGGGGCTCATTCCCATTTCATGCTGATCCCCGTCTTTTTGAATATCGGAACCAAGGAGATGAAGAAGGTACA
8  GAGATAGATACTCTACAGTTTCGGCTTCAGGTTCAGGTGCACTCGGAACCCCATGCTGCTAAAGATTCTCTGACCCCAA
9  TGAATCTATGTGAACCATAAAGTGTACACCAGGCACATGACATGGGTGCCCCCTGGGGAATCAGGCTGACCTCTTCCCAG
0  AGGGCACGATCAGACCAGTGCATGATGATATTCTCATCGCTCAGCTACGGCCTGGCCAGGAGATTGACTTACTCATGCAC
1  TGTGTCAAGGGCATTGGCAAGGATCATGCCAAGTTTTTCACCTGTGGCAACAGCCAGTTACAGGCTTCTGCCAGACATCAC
2  CCTGCTTGAGCCTGTGGAAGGGGATGCAGCGGAGGAGCTGAGGCGGTGCTTCTCGCCTGGTGTATTGAGGTGCAGGAAG
3  TCCAAGGTAAAAAGGTGGCCAGAGTTGCCAATCCCCGGCTAGATACCTTCAGCAGAGAAGTGTTCGGGAACGAGAAGCTA
4  AAGAAGGTTGTACGACTTGCTCGAGTTCGGGACCACTATATCTTTTCTGTGAGTCAACAGGGGTGTTGCCGCCAGAGGT
5  GCTAGTGAGTGAAGCCATCAAGGTACTGATGGGGAAGTGCCAACGGTTCTTGGATGAACTAGATGCAGTTCAGATGGACT
6  GAGGTTTGCATGGGACTTACCGGCATGGCTGTGTTTGGATACTCAAGTATCCTAGCTGAAGATTTTGGGTCCTGACTTGC
7  TGGTCACAGGACTGCTAGAGTCCAGTGTGACTGCAGGTCTGGGTTTTCTGAGACAATCCAGCCTTATTTTCAGTCAAT
8  ACATTTTCGTTAAATGTGCCACAAAATGTGACTTCCTGCCTTTGTAAGGCCTTAATGTAAATAAACTCACATCCAAATAAA
9  AAATCTTTTTTGCCAGAGACTA
0
```

Acinonyx jubatus



```
seqdump (8).txt  NM_203290.4.fa  seqdump (5).txt  seqdump (6).txt  seqdump (7).txt  seqdump (9).txt  seqdump (10).txt  seqd
1  >XM_042938873.1 PREDICTED: Panthera leo RNA polymerase I and III subunit C (POLR1C), transcript variant X1, mRNA
2  TGGGGGACGCATGCGCCAGTTTGTCTGCCCTAGTCACGTGGAGGGGTTGAAGATGGCGGCTGCTCAGGCGGTGGAAGAAAT
3  GCGGACCCGCGTGGTTCTAGGGGAGTTCGGGGTTCGCAATGTTTCATACAACCTGACTTTCCTGGTAACTATTTCGGGTTATG
4  ATGATGCCTGGGACCAAGACCGCTTCGAGAAGAATTTCCGTGTGGATGTAGTGCACCTGGATGAAAACCTCATTGGAGTTT
5  GACATGGTGGGAATTGATGCAGCCATTGCCAATGCTTTTCGGCGCATTTTATTAGCTGAGGTGCCAACCATGGCTGTGGA
6  AAAGGTCCTGGTGTACAACAACACGTCCATTGTCCAGGATGAGATCCTTGCTCATCGCTTGGGGCTCATTCCCATTTCATG
7  CTGATCCCCGTCTTTTTGAATATCGGAACCAAGGAGACGAAGAAGGTACAGAGATAGATACTCTACAGTTTTCGGCTTCAG
8  GTCAGGTGCACTCGGAACCCCAACGCTGCTAAAGATTCTCTGACCCCAATGAACTCTATGTGAACCATAAAGTGTACAC
9  CAGGCACATGACATGGGTGCCCCCTGGGGAATCAGGCTGACCTCTTCCCAGAGGGCACGATCAGACCAGTGCATGATGATA
0  TTCTCATCGCTCAGCTACGGCCTGGCCAGGAGATTGACTTACTCATGCACTGTGTCAAGGGCATTGGCAAGGATCATGCC
1  AAGTTTTTCACCTGTGGCAACAGCCAGTTACAGGCTTCTGCCAGACATCACCTTGCTTGAGCCTGTGGAAGGGGATGCAGC
2  GGAGGAGCTGAGGCGGTGCTTCTCGCCTGGTGTATTGAGGTGCAGGAAGTCCAAGGTAAAAAGGTGGCCAGAGTTGCCA
3  ATCCCCGGCTAGATACCTTCAGCAGAGAAGTGTTCGGGAACGAGAAGCTAAAGAAGGTTGTACGACTTGCTCGAGTTCGG
4  GACCACTATATCTTTTCTGTGAGTCAACAGGGGTGTTGCCACCAGAGGTGCTAGTGAGTGAAGCCATCAAGGTACTGAT
5  GGGGAAGTGCCAACGGTTCTTGGATGAACTAGATGCAGTTCAGATGGACTGAGGTTTGCATGGGACTTACCGGCATGGCT
6  GTGCTTGGATACTCAAGTATCCTAGCTGAAGATTTTGGGTCCTGACTTGCTGGTCACAGGACTGCTAGAGTCCAGTGTGA
7  CTGCAGGTCCTGGGTTTTCTGGGACAATCCCAGCCTTATTTTCAGTCAATACATTTTGTAAATGTGCCACAAAATGTGA
8  CTTCTGCCTTTGTAAGGCCTTAATGTAAATAAACTCACATCCAAATAAAAACTTTTGCCAGAGACTACTTCCGGAGTT
9  TTAGCCCCAATATGTGGGGCCTCACCTTAGAAGAGGTTTGGCATTATTTGGCACTT
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Panthera leo

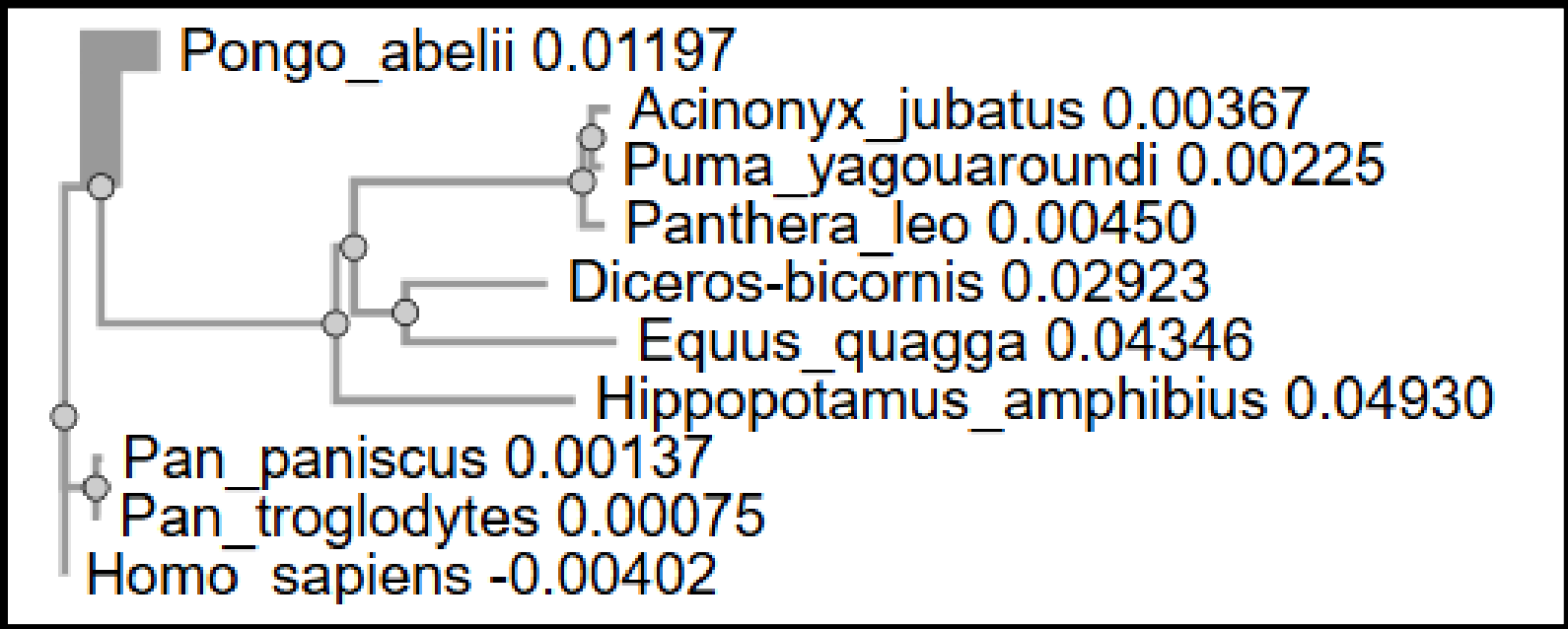


3. Make a MSA (Multiple Sequence Alignment) of the 10 sequences (including the one of your gene of interest). Can you obtain the percent identity matrix?

Clustal Omega

Multiple Sequence Alignment (MSA)

<https://www.ebi.ac.uk/jdispatcher/msa/clustalo>



```
#
# Percent Identity Matrix - created by Clustal2.1
#
#
```

1: Pongo_abelii	100.00	97.23	98.51	97.30	88.04	89.65	88.17	88.29	89.03	88.73
2: Pan_paniscus	97.23	100.00	99.61	99.79	88.25	89.40	88.54	88.51	88.71	87.35
3: Homo_sapiens	98.51	99.61	100.00	99.69	89.39	89.64	89.39	89.30	89.87	89.34
4: Pan_troglodytes	97.30	99.79	99.69	100.00	88.10	89.48	88.38	88.35	88.79	87.43
5: Acinonyx_jubatus	88.04	88.25	89.39	88.10	100.00	90.94	99.10	99.41	88.94	89.37
6: Diceros-bicornis	89.65	89.40	89.64	89.48	90.94	100.00	90.79	90.93	91.22	92.73
7: Panthera_leo	88.17	88.54	89.39	88.38	99.10	90.79	100.00	99.03	89.23	89.37
8: Puma_yagouaroundi	88.29	88.51	89.30	88.35	99.41	90.93	99.03	100.00	89.34	89.50
9: Hippopotamus_amphibius	89.03	88.71	89.87	88.79	88.94	91.22	89.23	89.34	100.00	89.47
10: Equus_quagga	88.73	87.35	89.34	87.43	89.37	92.73	89.37	89.50	89.47	100.00

4. For each nucleotide sequence, obtain the protein sequence (three should be from prediction and the remaining six, you should find theirs in databases)

From a database(NCBI) <https://www.ncbi.nlm.nih.gov>

Diceros bicornis

```
>ref|XP_058410273.1|:1-346 DNA-directed RNA polymerases I and III subunit RPAC1 isoform X1 [Diceros bicornis minor]
MAAAQAVEEMRSRVVLGEFGVRNVHTTDFPGNYSYDDAWDQDRFEKNFRVDVVHMDENSLEFDMVGIDA
AIAANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPRLFEYRNQGDEEGTEIDTLQF
RLQVRCTRNPAAKDSSDPNELYNHVKVYTRHMTWVPLGNQADLFPEGTIRPVHDDILIAQLRPGQEIDL
LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGDAAEELSRCFSPGVIELQEVQGKKVARVANPRL
DTFSREVFRHEKFKKVRLARVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRFLDELDAVQMD
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Hippopotamus amphibius kiboko

```
>ref|XP_057556662.1|:1-307 DNA-directed RNA polymerases I and III subunit RPAC1 isoform X3 [Hippopotamus amphibius kiboko]
MAAAQAVEEMRSRVVLGEFGIRNVHTTDFPGNYSYDDAWDQDRFEKNFRVDVVHMDENSLEFDMVGIDA
AIAANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIRADPRLFEYRNQGDEEGTEIDTLQF
RLQVRCTRNPAAKDSSDPNELYNHVKVYTRHMTWVPLGNQADVFPPEGTIRPVHDDILIAQLRPGQEIDL
LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGDAAEELSRCFSPGVIEVQEIQGKKVARVANPRL
DTFSREVFRNEKLKKVRLARVRNHYI
```

Equus quagga

```
>ref|XP_046495241.1|:1-346 DNA-directed RNA polymerases I and III subunit RPAC1 isoform X1 [Equus quagga]
MAAAQAVEEMRSRVVLGEFGVRNVHTTDFPGNYSYDDAWDQDHFEKNFRVDVVHMDENSLEFDMVGIDA
AIAANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPRLFEYRNQGDEEGTEIDTLQF
RLQVRCTRNPAAKDSSDPSELYNVHVKVYTRHMIWVPLGNQADLFPEGTIRPVHDDILIAQLRPGQEIDL
LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGDAAEELSRCFSPGVIEVQEVQGKKVARVANPRL
DTFSREVFRHEKLKKVRLARVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRFLDELEAVQMD
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Panthera leo

>ref|XP_042794807.1|:1-346 DNA-directed RNA polymerases I and III subunit RPAC1 isoform X1 [Panthera leo]

MAAAQAVEEMRTRVVLGEFGVRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHLDENSLEFDMVGIDA
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RLQVRCTRNPAAKDSSDPNELYNHKVYTRHMTWVPLGNQADLFPEGTIRPVHDDILIAQLRPGQEIDL
LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGDAAEELRRCFSPGVIEVQEVQGKKVARVANPRL
DTFSREVERNEKLKKVVRLARVRDHYIFSVESTGVLPPPEVLVSEAIKVLMGKCQRFLDELDAVQMD

Puma yagouroundi

>ref|XP_040329548.1|:1-346 DNA-directed RNA polymerases I and III subunit RPAC1 isoform X1 [Puma yagouaroundi]

MAAAQAVEEMRTRVVLGEFGVRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHLDENSLEFDMVGIDA
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RLHVRCTRNPAAKDSSDPNELYNHKVYTRHMTWVPLGNQADLFPEGTIRPVHDDILIAQLRPGQEIDL
LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGDAAEELRRCFSPGVIEVQEVQGKKVARVANPRL
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Acinonyx jubatus

>ref|XP_014926702.1|:1-346 DNA-directed RNA polymerases I and III subunit RPAC1 isoform X1 [Acinonyx jubatus]

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RLQVRCTRNPAAKDSSDPNELYNHKVYTRHMTWVPLGNQADLFPEGTIRPVHDDILIAQLRPGQEIDL
LMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGDAAEELRRCFSPGVIEVQEVQGKKVARVANPRL
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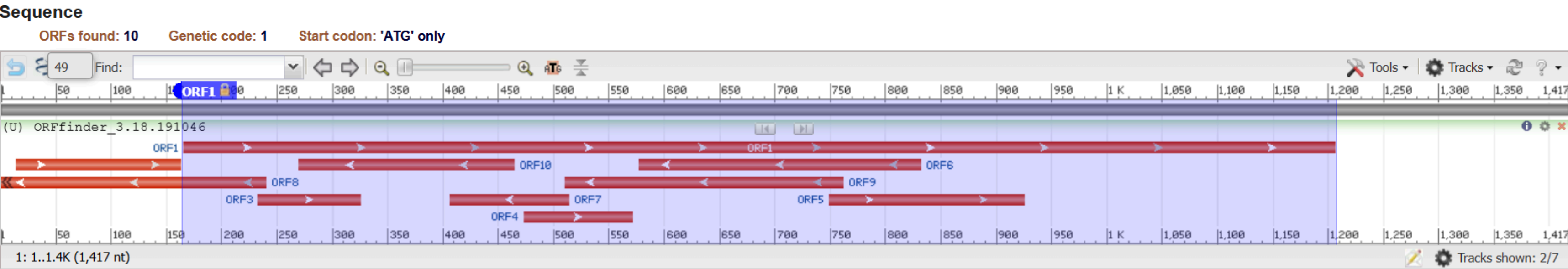
From prediction:

<https://www.ncbi.nlm.nih.gov/orffinder/>



Pan troglodytes

Open Reading Frame Viewer



Six-frame translation...

ORF1 (346 aa) Display ORF as... Mark

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>lcl|ORF1
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VQDEILAHRLGLIPIHADPRLFYRNQGDDEGTEIDTLQFRLQVRCTRNP
HAAKDSSDPNELVYVNHKVYTRHMTWIPLGNQADLFPEGTIRPVHDDILIA
QLRPGQEIDLLMHCVKGIGKDHAKFSPVATASYRLLPDITLLEPVEGEAA
EELSRCFSPGVIEVQEVQKKVARVANPRLDTSREIFRNEKLKKVRLA
RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRLDELDAVQMD
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ORF1

SmartBLAST

BLAST

Marked set (0)

SmartBLAST best hit titles... ?

BLAST

Mark subset... Marked: 0 Download marked set as Protein FASTA ▾

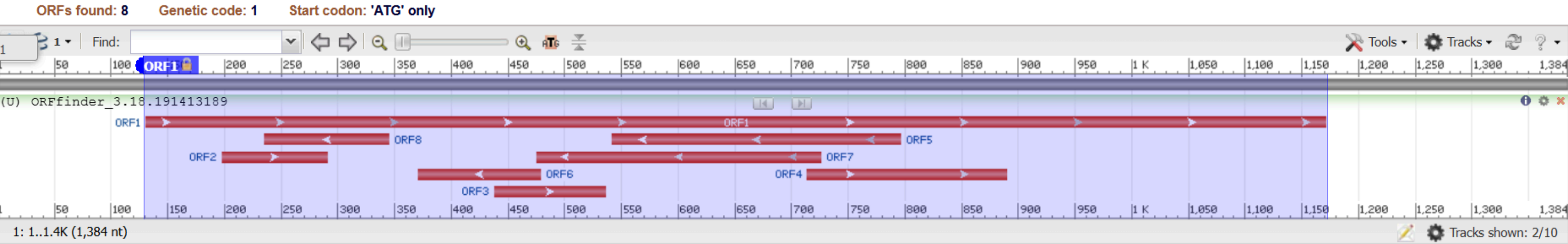
Label	Strand	Frame	Start	Stop	Length (nt aa)
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ORF6	-	2	831	577	255 84
ORF9	-	3	761	510	252 83
ORF8	-	2	240	>1	240 79
ORF10	-	3	464	270	195 64
ORF5	+	2	749	925	177 58
ORF2	+	2	14	163	150 49
ORF7	-	2	513	406	108 35
ORF4	+	2	473	571	99 32
ORF3	+	2	233	325	93 30



Pongo abelii

Open Reading Frame Viewer

Sequence



[Six-frame translation...](#)

ORF1 (346 aa) [Display ORF as...](#) [Mark](#)

>lcl|ORF1
MAASQAVEEMRSRVVLGEFGVRNVHTTDFPGNYSYDDAWDQDRFEKNFR
VDVVHMDENSLEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSI
VQDEILAHRLGLPIHADPRLFEYRNQDDEEGTEIDTLQFRLQVRCTRN
HAAKSSDPNELVYVNHKVYTRHMTWIPLGNQADLFPEGTIRPVHNDILIA
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EELSRCFSPGVIEVQEVQGGKVARVANPRLDTSREIFRNEKLKKVRLA
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ORF1

[SmartBLAST](#)

[BLAST](#)

Marked set (0)

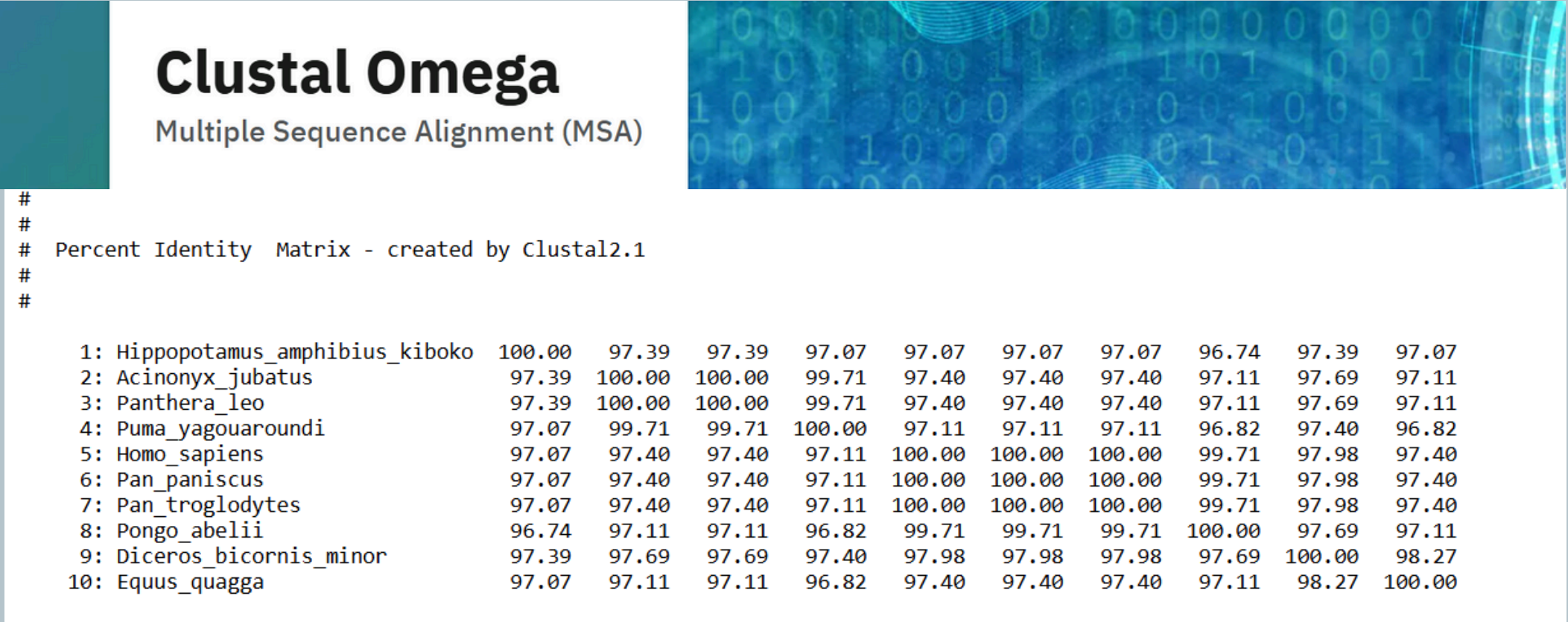
SmartBLAST best hit titles... [?](#)

[BLAST](#)

[Mark subset...](#) Marked: 0 [Download marked set](#) as [Protein FASTA](#) [v](#)

Label	Strand	Frame	Start	Stop	Length (nt aa)
ORF1	+	2	131	1171	1041 346
ORF5	-	1	796	542	255 84
ORF7	-	2	726	475	252 83
ORF4	+	3	714	890	177 58
ORF8	-	2	345	235	111 36
ORF6	-	1	478	371	108 35
ORF3	+	3	438	536	99 32
ORF2	+	3	198	290	93 30

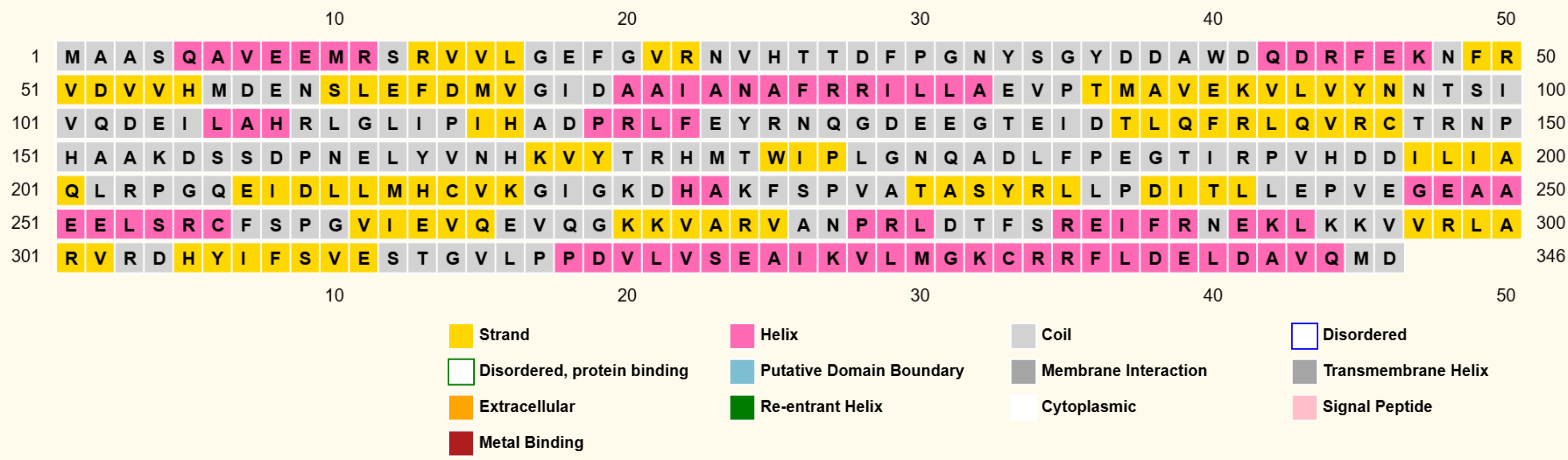
5. Perform the MSA again but with the protein sequences, can you see conserved regions? Are any of these conserved regions present in your gene of interest?



- According to the multiple sequence alignment, the most part of the protein is conserved all along the different species I selected.
- In contrast with the homo sapiens gene of interest, the regions are also highly conserved.

Hippopotamus_amphibius_kiboko	MAAAQAVEEMRTAVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHMDENS	60
Acinonyx_jubatus	MAAAQAVEEMRTRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHLDENS	60
Panthera_leo	MAAAQAVEEMRTRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHLDENS	60
Puma_yagouaroundi	MAAAQAVEEMRTRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHLDENS	60
Homo_sapiens	MAASQAVEEMRSRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHMDENS	60
Pan_paniscus	MAASQAVEEMRSRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHMDENS	60
Pan_troglodytes	MAASQAVEEMRSRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHMDENS	60
Pongo_abelii	MAASQAVEEMRSRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHMDENS	60
Diceros_bicornis_minor	MAAAQAVEEMRSRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDRFEKNFRVDVVHMDENS	60
Equus_quagga	MAAAQAVEEMRSRVVLGEGFGRNVHTTDFPGNYSGYDDAWDQDHFEKNFRVDVVHMDENS	60
	;**;*****;*****;*****;*****;*****;*****;*****;*****	
Hippopotamus_amphibius_kiboko	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIRADPR	120
Acinonyx_jubatus	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
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Puma_yagouaroundi	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
Homo_sapiens	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
Pan_paniscus	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
Pan_troglodytes	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
Pongo_abelii	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
Diceros_bicornis_minor	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
Equus_quagga	LEFDMVGIDAAIANAFRRILLAEVPTMAVEKVLVYNNTSIVQDEILAHRLGLIPIHADPR	120
	*****;*****;*****;*****;*****;*****;*****;*****;*****;*****	
Hippopotamus_amphibius_kiboko	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWVPLGN	180
Acinonyx_jubatus	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWVPLGN	180
Panthera_leo	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWVPLGN	180
Puma_yagouaroundi	LFEYRNQGDDEEGTEIDTLQFRLHVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWVPLGN	180
Homo_sapiens	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWIPLGN	180
Pan_paniscus	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWIPLGN	180
Pan_troglodytes	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWIPLGN	180
Pongo_abelii	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWIPLGN	180
Diceros_bicornis_minor	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPNELVYNHKVYTRHMTWVPLGN	180
Equus_quagga	LFEYRNQGDDEEGTEIDTLQFRLQVRCTRNPAAKDSSDPSELVYNHKVYTRHMIWVPLGN	180
	*****;*****;*****;*****;*****;*****;*****;*****;*****;*****	
Hippopotamus_amphibius_kiboko	QADVPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Acinonyx_jubatus	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Panthera_leo	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Puma_yagouaroundi	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Homo_sapiens	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Pan_paniscus	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Pan_troglodytes	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Pongo_abelii	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Diceros_bicornis_minor	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
Equus_quagga	QADLFPEGTIRPVHDDILIAQLRPGQEQIDLLMHCVKGIGKDHAKEFSPVATASYRLLPDIT	240
	*****;*****;*****;*****;*****;*****;*****;*****;*****;*****	
Hippopotamus_amphibius_kiboko	LLEPVEGDAAEELSRCFSPGVIEVQEQGKKVARVANPRLDTFSREVFNRNEKLKKVVRLA	300
Acinonyx_jubatus	LLEPVEGDAAEELRRCFSPGVIEVQEQGKKVARVANPRLDTFSREVFNRNEKLKKVVRLA	300
Panthera_leo	LLEPVEGDAAEELRRCFSPGVIEVQEQGKKVARVANPRLDTFSREVFNRNEKLKKVVRLA	300
Puma_yagouaroundi	LLEPVEGDAAEELRRCFSPGVIEVQEQGKKVARVANPRLDTFSREVFNRNEKLKKVVRLA	300
Homo_sapiens	LLEPVEGEAAEELSRCFSPGVIEVQEQGKKVARVANPRLDTFSREIFRNEKLKKVVRLA	300
Pan_paniscus	LLEPVEGEAAEELSRCFSPGVIEVQEQGKKVARVANPRLDTFSREIFRNEKLKKVVRLA	300
Pan_troglodytes	LLEPVEGEAAEELSRCFSPGVIEVQEQGKKVARVANPRLDTFSREIFRNEKLKKVVRLA	300
Pongo_abelii	LLEPVEGEAAEELSRCFSPGVIEVQEQGKKVARVANPRLDTFSREIFRNEKLKKVVRLA	300
Diceros_bicornis_minor	LLEPVEGDAAEELSRCFSPGVIELQEQGKKVARVANPRLDTFSREVFNRNEKLKKVVRLA	300
Equus_quagga	LLEPVEGDAAEELSRCFSPGVIEVQEQGKKVARVANPRLDTFSREVFNRNEKLKKVVRLA	300
	*****;*****;*****;*****;*****;*****;*****;*****;*****;*****	
Hippopotamus_amphibius_kiboko	RVRNHYI-----	307
Acinonyx_jubatus	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCQRFLDELDAVQMD	346
Panthera_leo	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCQRFLDELDAVQMD	346
Puma_yagouaroundi	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCQRFLDELDAVQMD	346
Homo_sapiens	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRFLDELDAVQMD	346
Pan_paniscus	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRFLDELDAVQMD	346
Pan_troglodytes	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRFLDELDAVQMD	346
Pongo_abelii	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRFLDELDAVQMD	346
Diceros_bicornis_minor	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRFLDELDAVQMD	346
Equus_quagga	RVRDHYIFSVESTGVLPPDVLVSEAIKVLMGKCRRFLDELEAVQMD	346
	;	

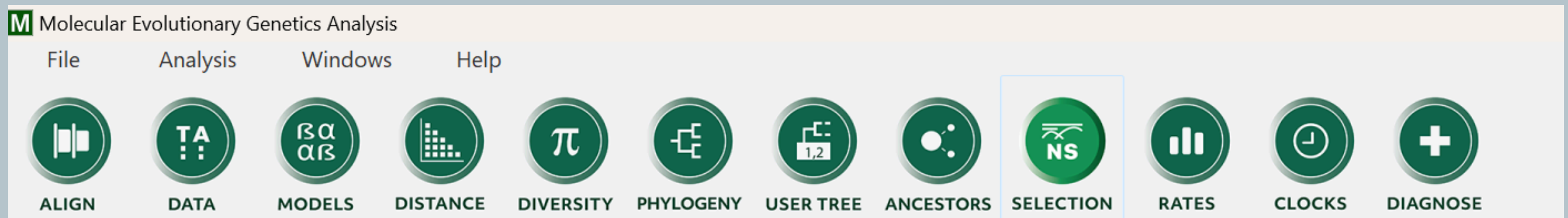
6. Perform a prediction of secondary structure of your original gene of interest (use the protein sequence for this). Have these conserved regions any secondary structure in particular? Can you think of a reason?



As the protein is highly conserved in all of the different species, we can deduce that the protein has a very robust structure that does not allow for big variations in its sequence or structure. This means that the protein has a primordial and specific function all over the different species.

7. Perform a Maximum Likelihood Tree with Bootstrap (100) of the mammal sequences (add an outgroup, i.e. an orthologue plant sequence). Plot the tree in circular view

With our different sequences we can make a phylogenetic tree that tells us how related the different sequences from the gene of interest in human and the different species selected are. We are going to use **MEGA** (**M**olecular **E**volutionary **G**enetics **A**nalysis)



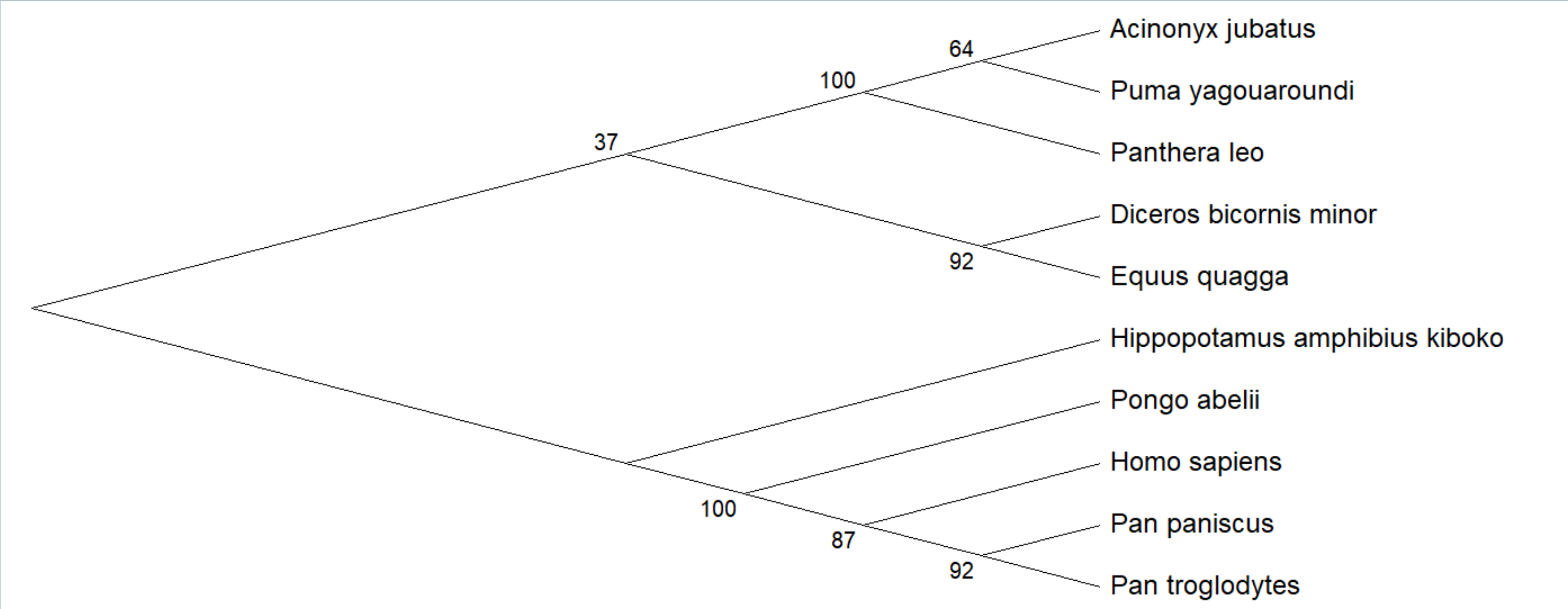
We first introduce our sequences in the app, align them, and search for the best model which applies to the sum of our sequences.

[illegible]

Table. Maximum Likelihood analysis of substitution models																						
Model	Parameters	BIC	AICc	<i>lnL</i>	(+I)	(+G)	<i>R</i>	<i>f</i> (A)	<i>f</i> (T)	<i>f</i> (C)	<i>f</i> (G)	<i>r</i> (AT)	<i>r</i> (AC)	<i>r</i> (AG)	<i>r</i> (TA)	<i>r</i> (TC)	<i>r</i> (TG)	<i>r</i> (CA)	<i>r</i> (CT)	<i>r</i> (CG)	<i>r</i> (GA)	<i>r</i> (GT)
K2+I	19	8902.826	8761.484	-4361.712	0.53	n/a	2.83	0.250	0.250	0.250	0.250	0.033	0.033	0.185	0.033	0.185	0.033	0.033	0.185	0.033	0.185	0.033
K2+G+I	20	8907.080	8758.302	-4359.118	0.53	2.52	2.92	0.250	0.250	0.250	0.250	0.032	0.032	0.186	0.032	0.186	0.032	0.032	0.186	0.032	0.186	0.032
T92+I	20	8917.282	8768.504	-4364.218	0.53	n/a	2.83	0.248	0.248	0.252	0.252	0.032	0.033	0.186	0.032	0.186	0.033	0.032	0.183	0.033	0.183	0.032
T92+G+I	21	8921.519	8765.305	-4361.616	0.53	2.51	2.92	0.248	0.248	0.252	0.252	0.032	0.032	0.188	0.032	0.188	0.032	0.032	0.185	0.032	0.185	0.032
HKY+I	22	8930.363	8766.714	-4361.317	0.53	n/a	2.84	0.249	0.247	0.228	0.275	0.032	0.030	0.203	0.033	0.169	0.036	0.033	0.182	0.036	0.184	0.032
TN93+I	23	8933.851	8762.767	-4358.340	0.53	n/a	2.83	0.249	0.247	0.228	0.275	0.032	0.030	0.168	0.032	0.202	0.036	0.032	0.218	0.036	0.152	0.032
HKY+G+I	23	8934.526	8763.442	-4358.677	0.53	2.48	2.94	0.249	0.247	0.228	0.275	0.031	0.029	0.205	0.032	0.170	0.035	0.032	0.184	0.035	0.186	0.032

Once we have found the best model(in this case, Kimura), we can perform the Maximum Likelihood Tree, with a bootstrap of 1000.

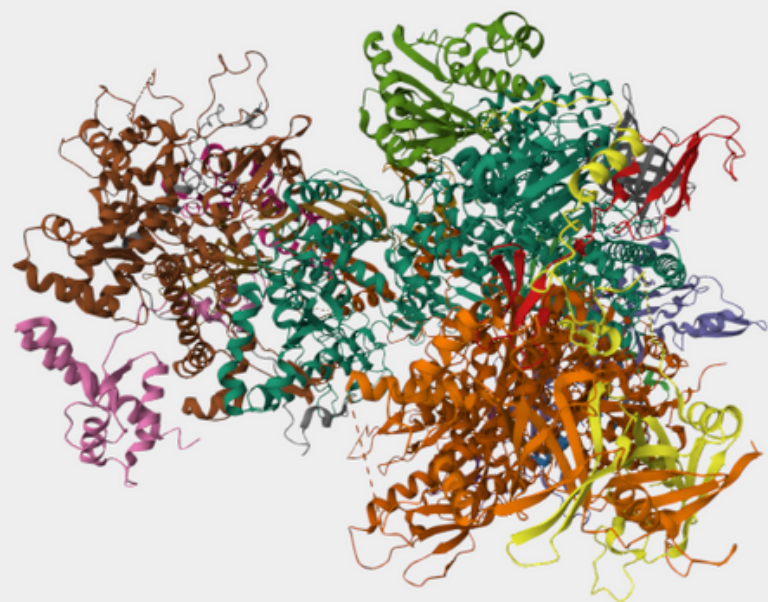
LINEAR VIEW



CIRCULAR VIEW



Does your gene have a solved protein structure? If so, retrieve the identifier. Can you tell which method was used to obtain this protein structure? If it has not, where have you found that information?



O15160 · RPAC1_HUMAN

Protein ⁱ	DNA-directed RNA polymerases I and III subunit RPAC1	Amino acids	346 (go to sequence)
Gene ⁱ	POLR1C	Protein existence ⁱ	Evidence at protein level
Status ⁱ	 UniProtKB reviewed (Swiss-Prot)	Annotation score ⁱ	 5/5
Organism ⁱ	Homo sapiens (Human)		

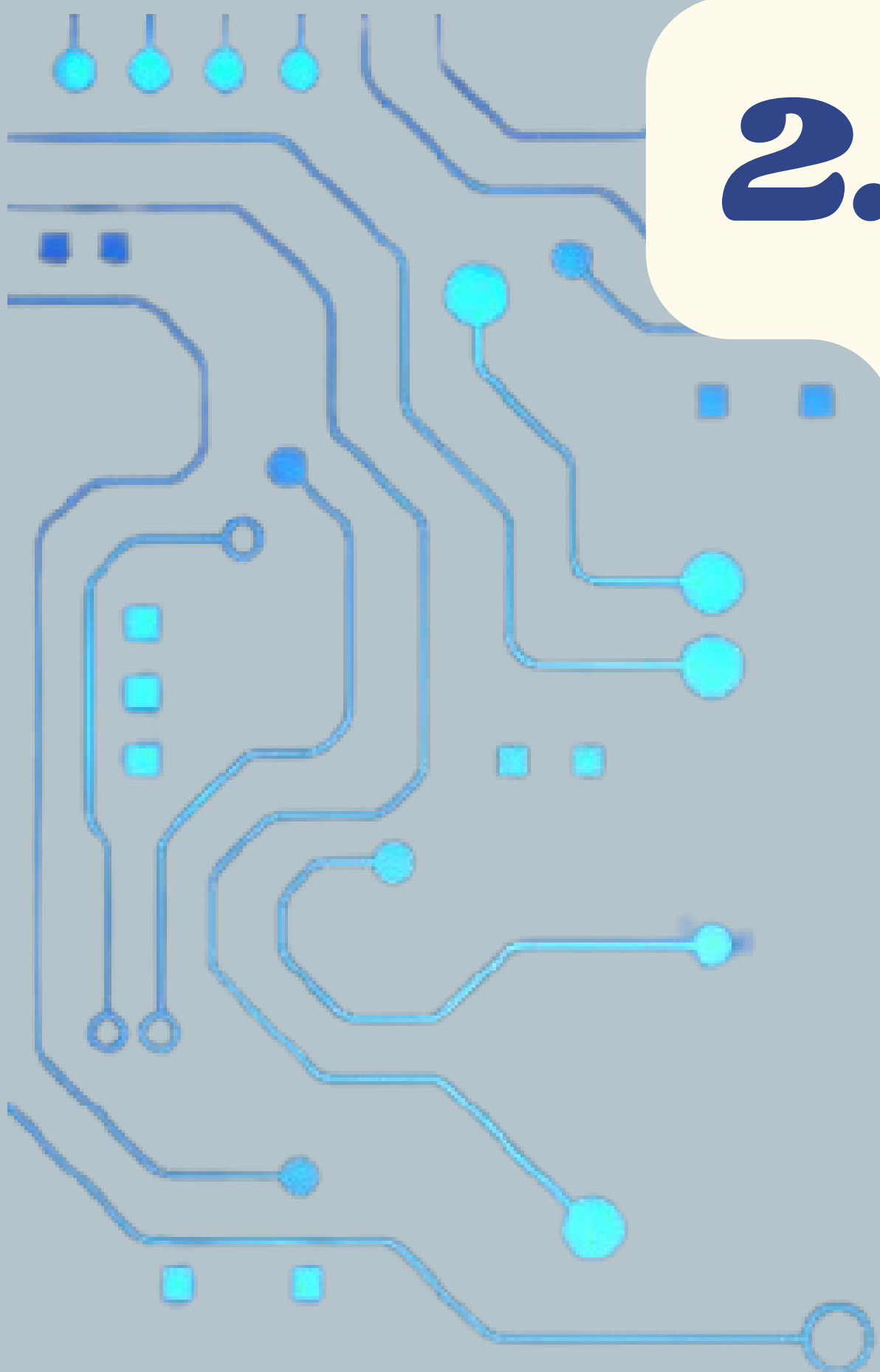
PDB	7OB9	EM	2.70 Å	C	1-346
-----	------	----	--------	---	-------

IDENTIFIER:

-UNIPROT: O15160

-NCBI: NP_001305805.1

- According to UniProt, there is evidence of the existence of the protein at experimental level. This means that this protein has been, by laboratory methods, proved to exist. These laboratory methods can range from a direct sequencing of the protein by mass spectrometry, to crystallography or magnetic resonance.
- In this protein in particular, the structure has been proved by electronic microscopy.



2. ASSEMBLY AND ANNOTATION

```
TGTTGCTTCGGCGGGCGCCCGCAAGGGTGCCCG  
GGCCTGCCGTGGCAGATCCCCAACGCCGGGGCC  
TCTCTTGGCTCCAGCATCGATGAAGAACGCAG  
CAGCATCGATGAAGAACGCAGCGAAACGCGAT  
CGATACTTCTGAGTGTTCTTAGCGAACTGTCA  
CGGATCTCTTGGCTCCAGCATCGATGAAGAAC  
ACAACGGATCTCTTGGCTCCAGCATCGATGAA  
CGGATCTCTTGGCTCCAGCATCGATGAAGAAC  
GATGAAGAACGCAGCGAAACGCGATATGTAAT
```

1. What type of input reads do you have and which sequencing technology?

Both SRR3137854-forward.fastqsanger and SRR31378544-reverse.fastqsanger are FASTQ files, and the technology used to obtain them is probably **Illumina**, looking at the headers and the format.

The other file, SRR21079301.fastq, is also a FASTQ file, but in this case the technology used for it was a long-read sequencing platform, such as **Oxford Nanopore technology**.

2. Load the files and evaluate the quality regarding the type of the file.

IN ILLUMINA

FastQC Read Quality reports (Galaxy Version 0.74+galaxy1)

We recommend using **Falco** instead. Falco is 3 times faster and offers the same functionality as FASTQC.

Tool Parameters

Raw read data from your current history *

accepted formats ▼ switch to column select ▼

2: SRR3137854-forward.fastqsanger x 1: SRR31378544-reverse.fastqsanger x

! This is a batch mode input field. Individual jobs will be triggered for each dataset.

Contaminant list - optional

We use FASTQC to analyze the quality of the reads

We later use MULTIQC to join both the reverse read and the forward read

The screenshot displays the MultiQC web interface. At the top, a header bar contains the MultiQC logo, a description 'aggregate results from bioinformatics analyses into a single report (Galaxy Version 1.27+galaxy3)', and icons for star, print, and download, along with a 'Run Tool' button. Below the header, the 'Tool Parameters' section is visible. Underneath, the 'Results' section is expanded, showing a list of results. The first result, '1: Results', is selected and expanded. It contains a dropdown menu for 'Which tool was used generate logs?' with 'FastQC' selected. Below this, the 'FastQC output' section is expanded, showing a dropdown for 'Type of FastQC output?' with 'Raw data' selected. At the bottom, the 'FastQC output' section shows a list of output files: '7: FastQC on data 2: RawData' and '5: FastQC on data 1: RawData'. There are also icons for accepted formats and a 'switch to column select' option.

In general, the quality is good: It has a high quality score, the GCs percentage is balanced between them, there is no adapter contamination, no overrepresented sequences... Although, the general percentage of GC content is high and the duplication rate, in the case of the forward read the most, could be problematic as it is more than a quarter of the sequence.

Sample Name

SRR3137854-forward_fastqsanger

SRR31378544-reverse_fastqsanger

Dups

26.2 %

21.9 %

GC

57.0 %

57.0 %

Seqs

0.7 M

0.7 M

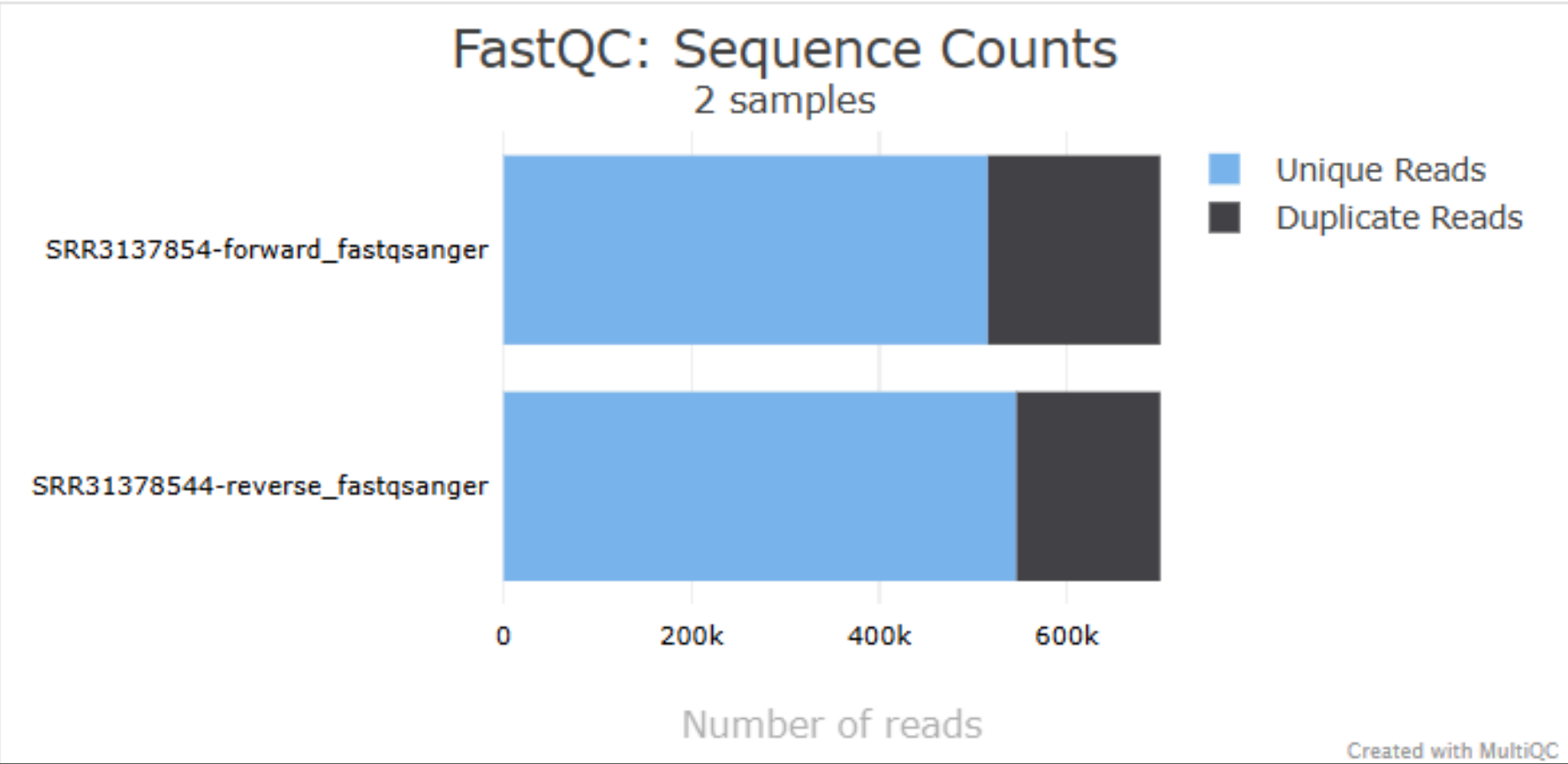
Sequence Counts

Sequence counts for each sample. Duplicate read counts are an estimate only.

Percentages

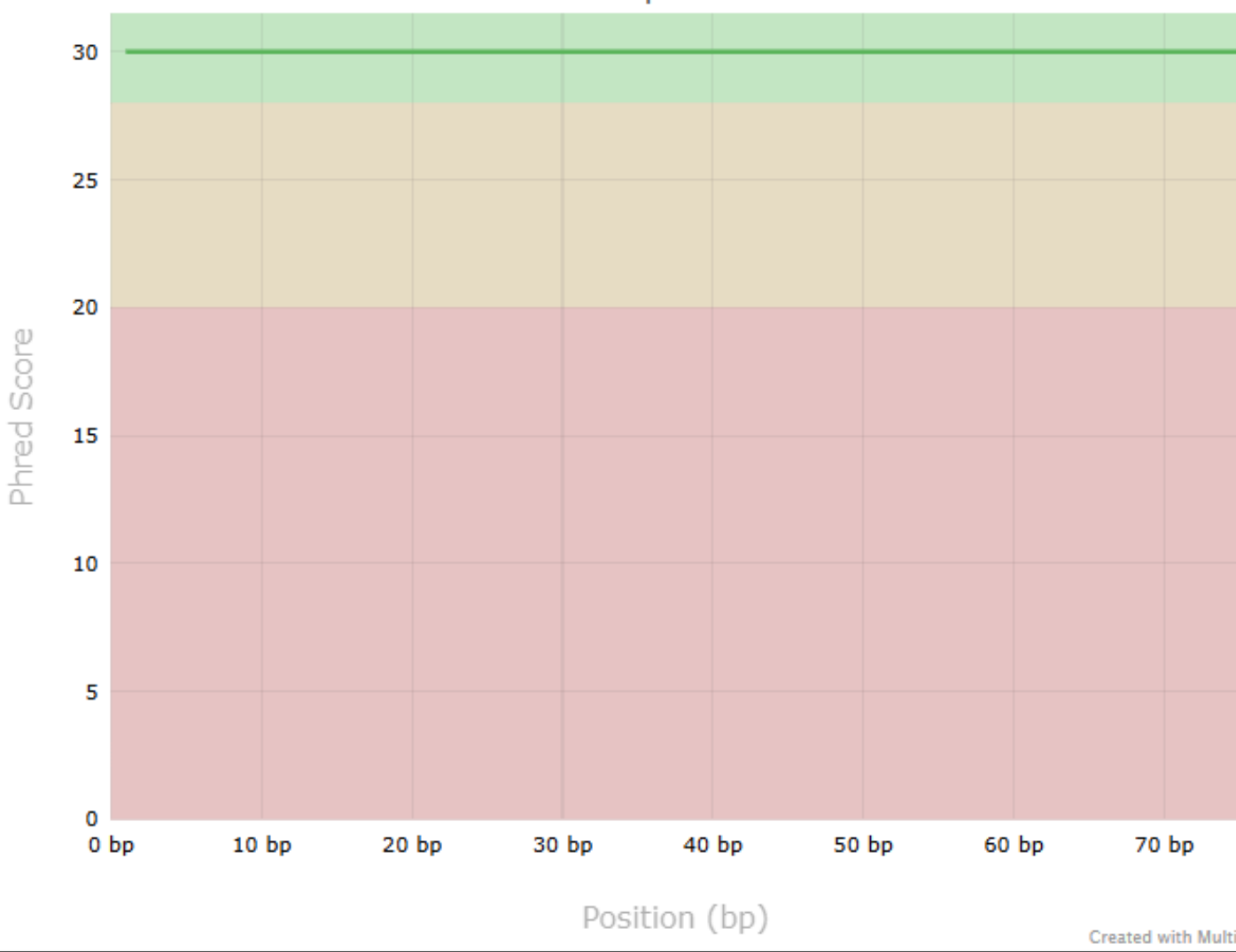
Summarize plot

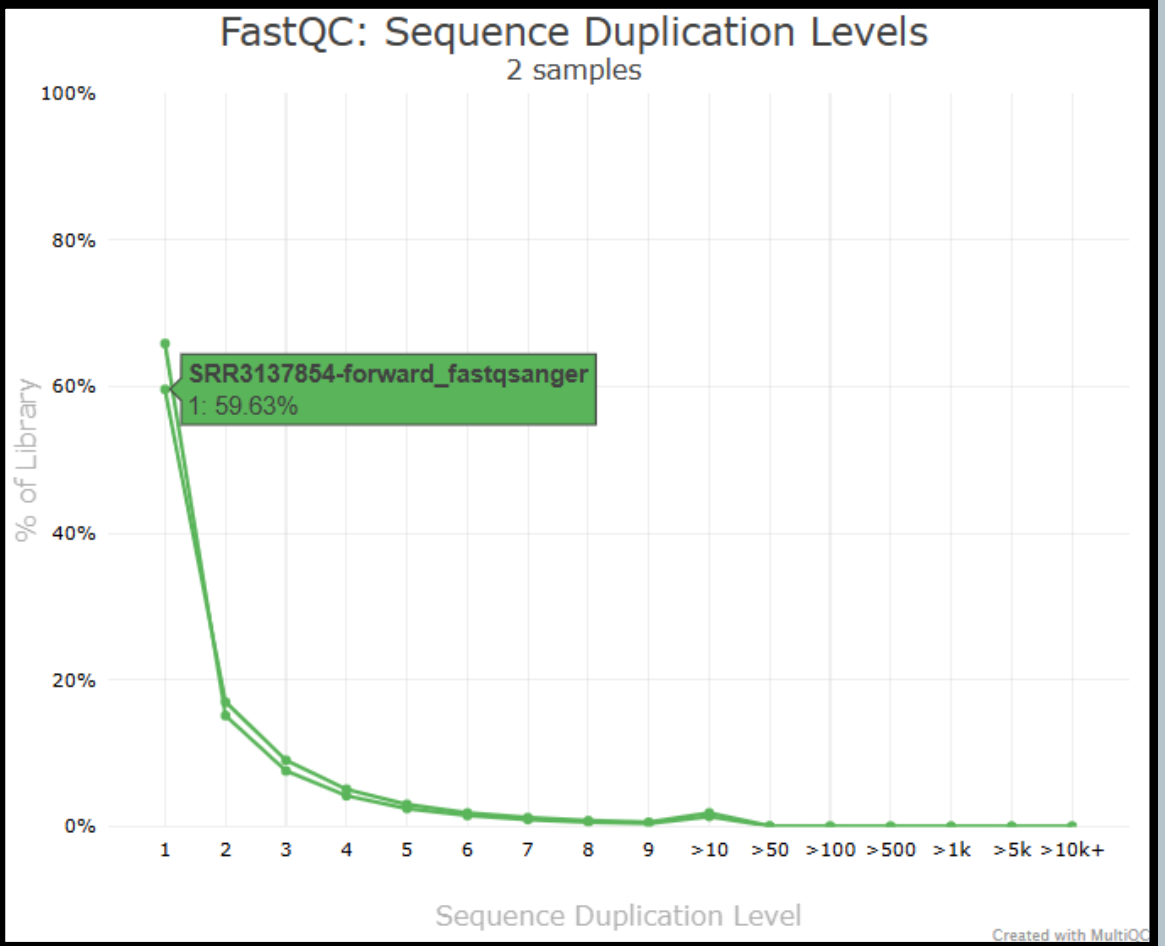
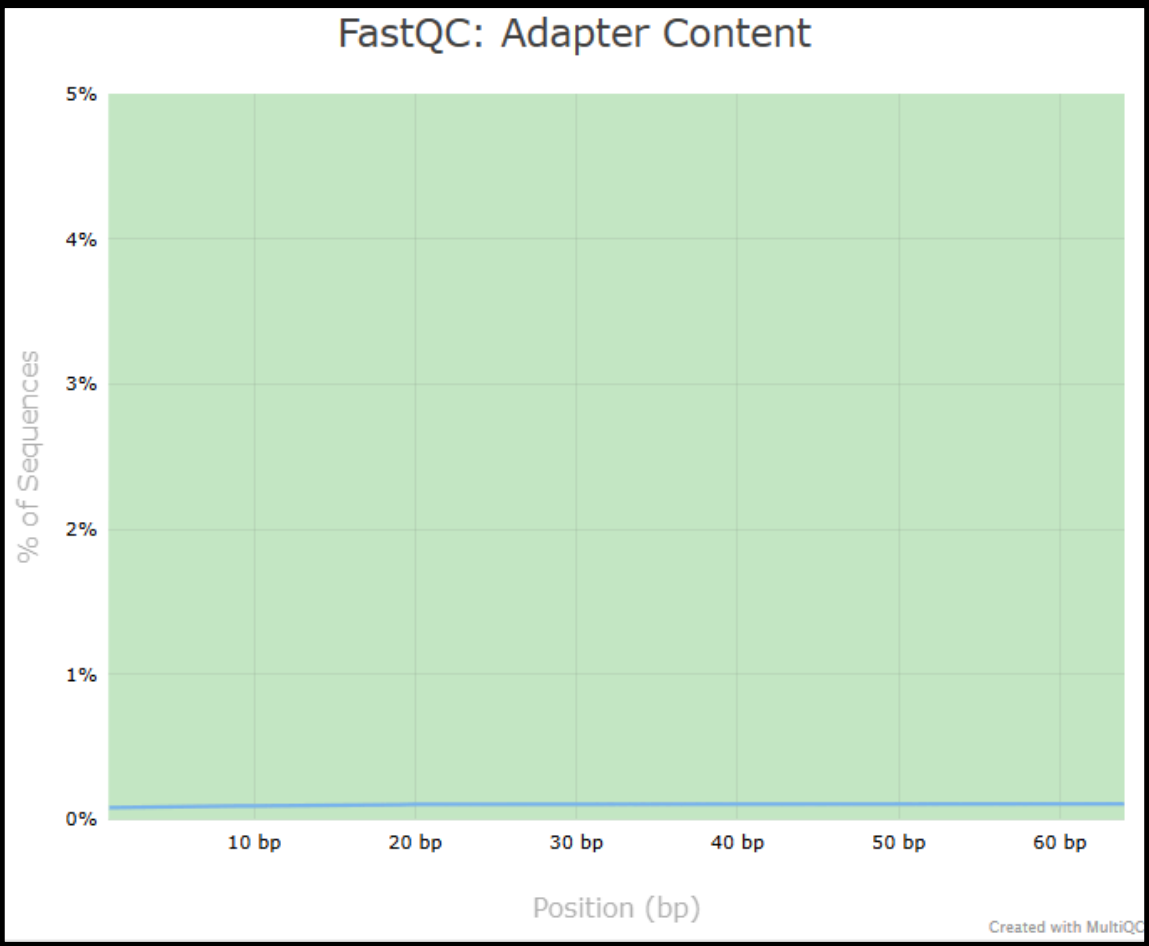
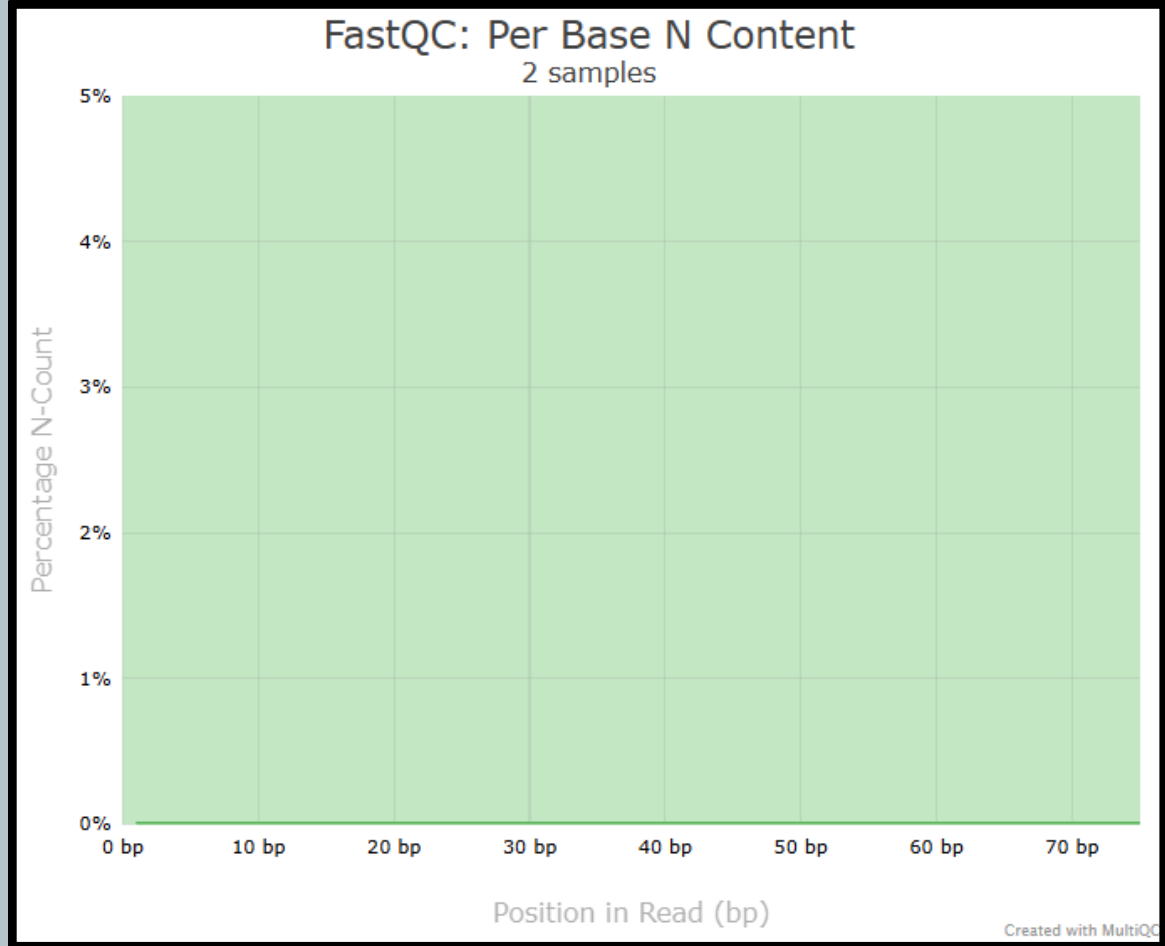
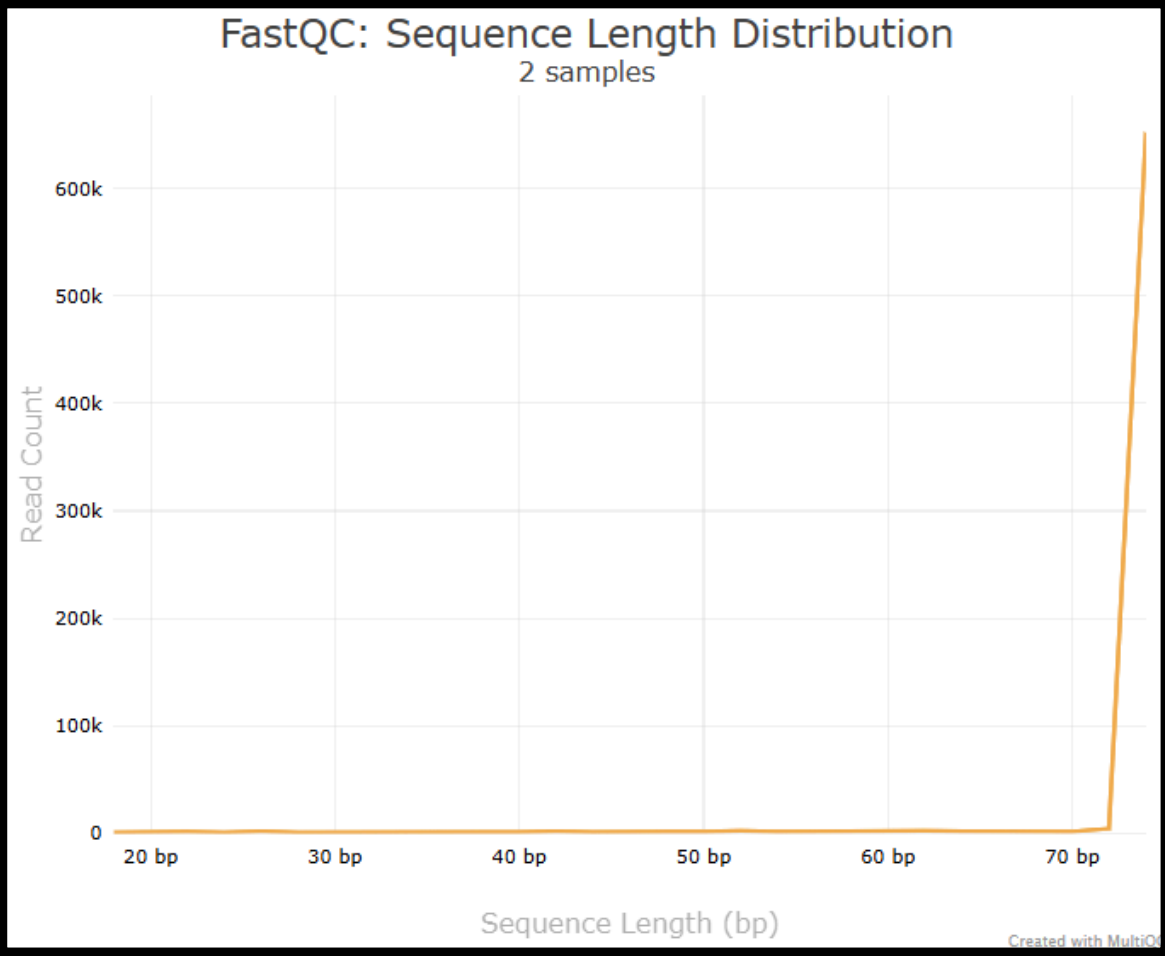
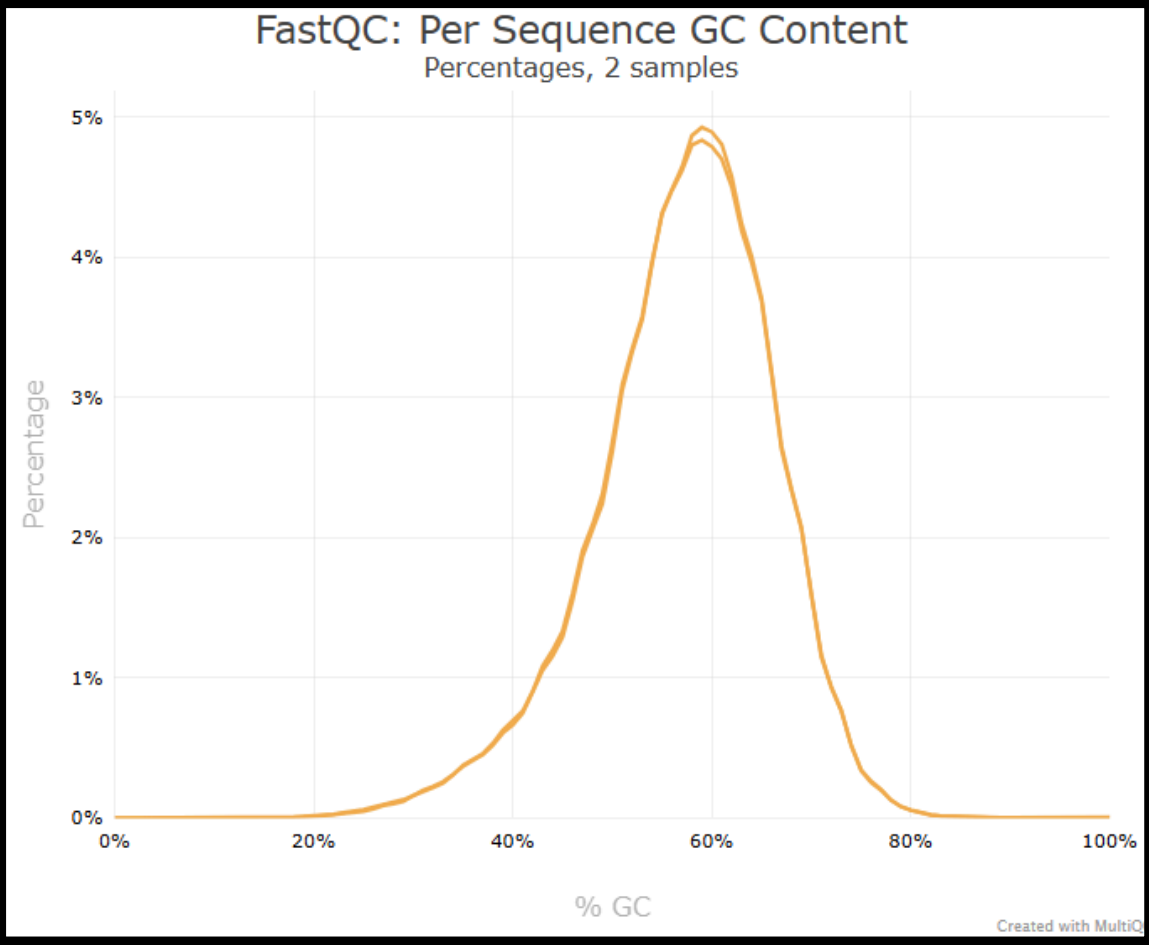
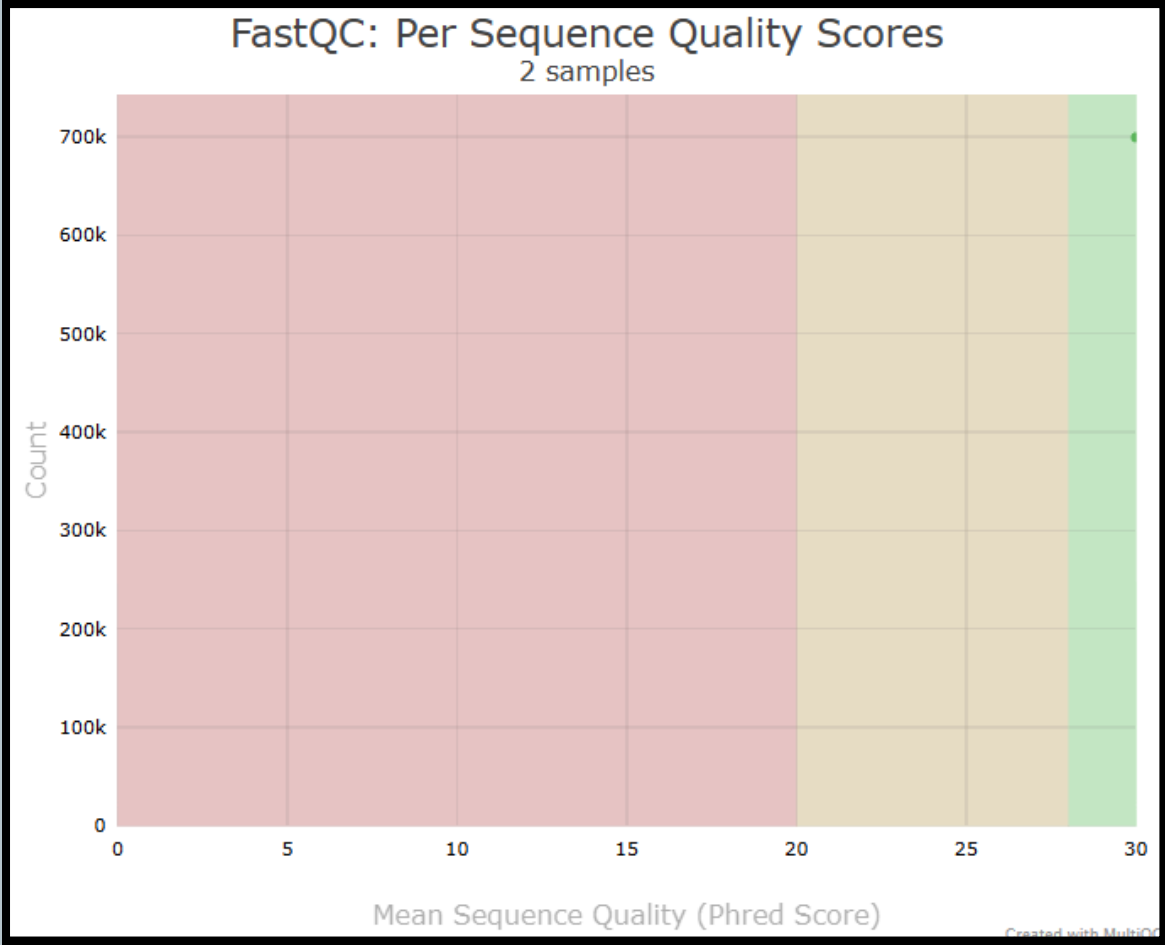
Export...



FastQC: Mean Quality Scores

2 samples



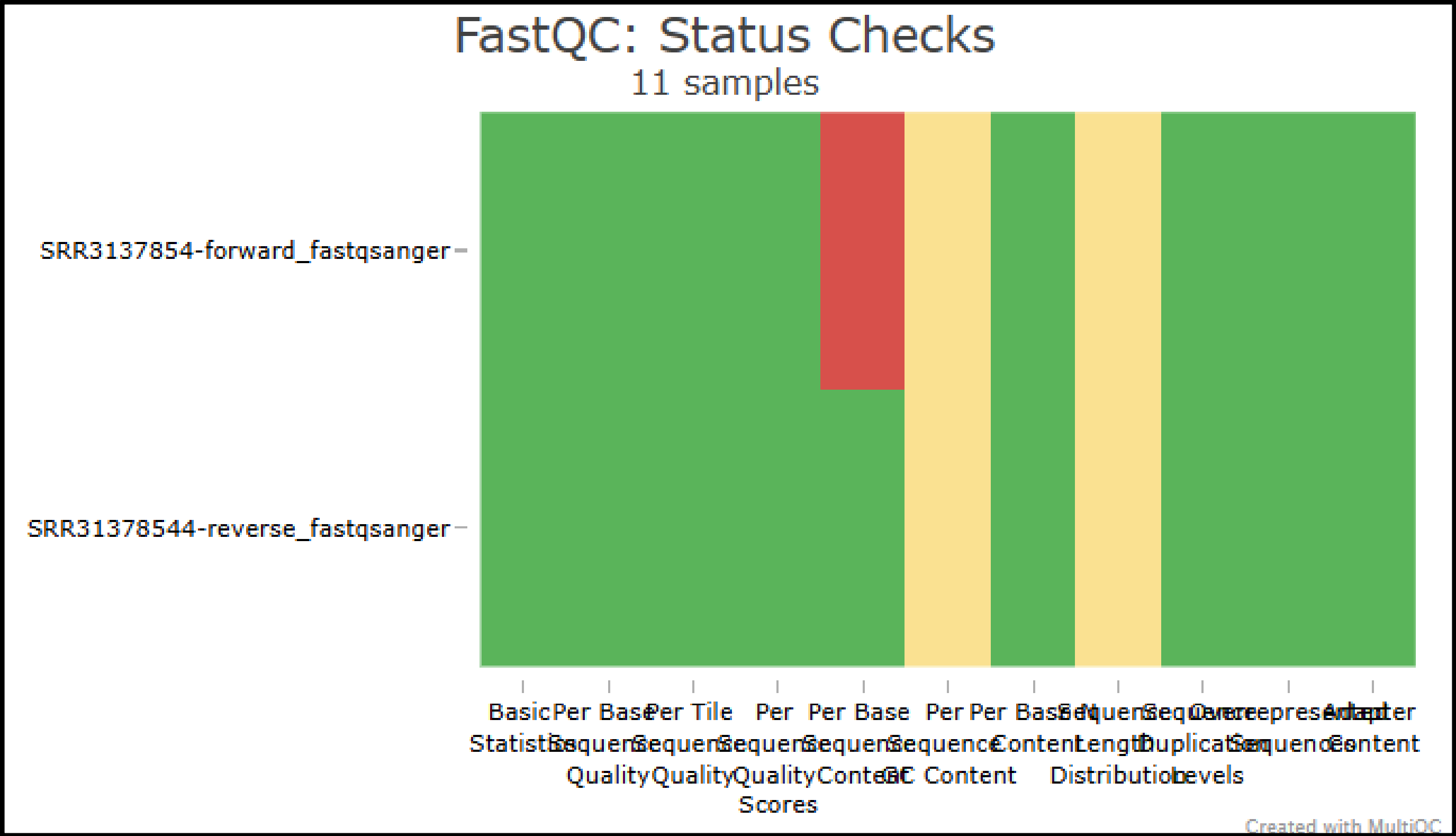


Overrepresented sequences by sample

Help

The total amount of overrepresented sequences found in each library.

2 samples had less than 1% of reads made up of overrepresented sequences



IN NANOPORE

 **NanoPlot** Plotting suite for Oxford Nanopore sequencing data and alignments (Galaxy Version 1.44.1+galaxy0)



Run Tool

Tool Parameters

Select multifile mode

batch

When supplying multiple files, batch mode will run NanoPlot on each file separately, while combined mode will run NanoPlot once on all the reads together. When only supplying a single input file, this setting does not matter.

Type of the file(s) to work on

fastq

files *



3: SRR21079301.fastq

accepted formats ▼

(--fastq)

According to Nanoplot, the quality is good, as:

- The mean and median quality is 30, having all of the reads over the Q10, although none of them over the Q30 (which are extremely high quality).
- The mean read length is 8744, which is high, and the median read is 9228 which is also high.
- The N50 is 11047= half of the data are reads that are longer than this number,proving to be long reads.

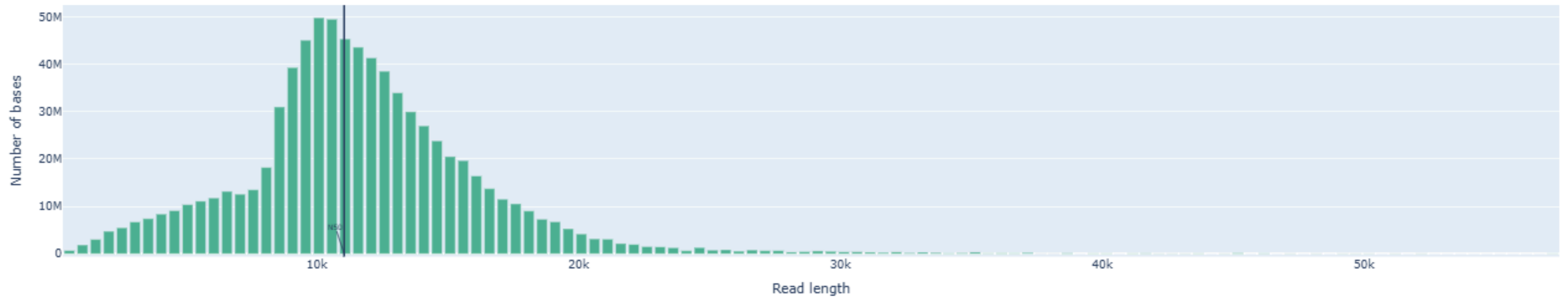
As we can see, the quality of the reads seems good. I also calculated the coverage to make sure it was correct (optimal value= 50-100X).

COBERTURA= $\frac{\text{Número de bases secuenciadas}}{\text{Tamaño del genoma}}$

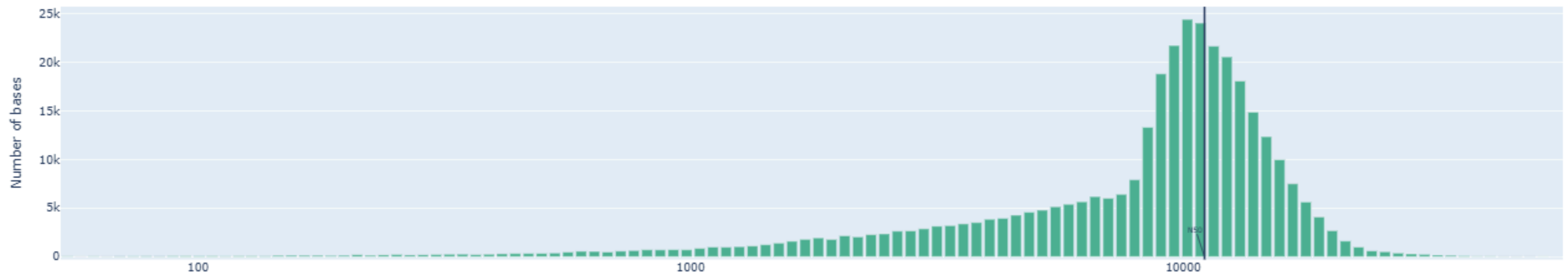
COBERTURA= $\frac{790.664.855 \text{ pb}}{5.3-5.5 \text{ Mb}} = \sim 146.4X$

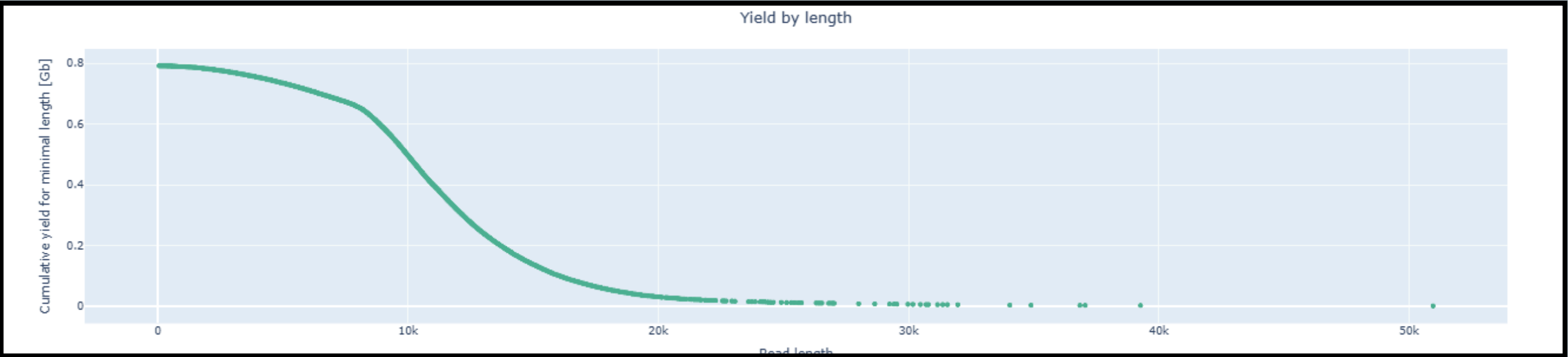
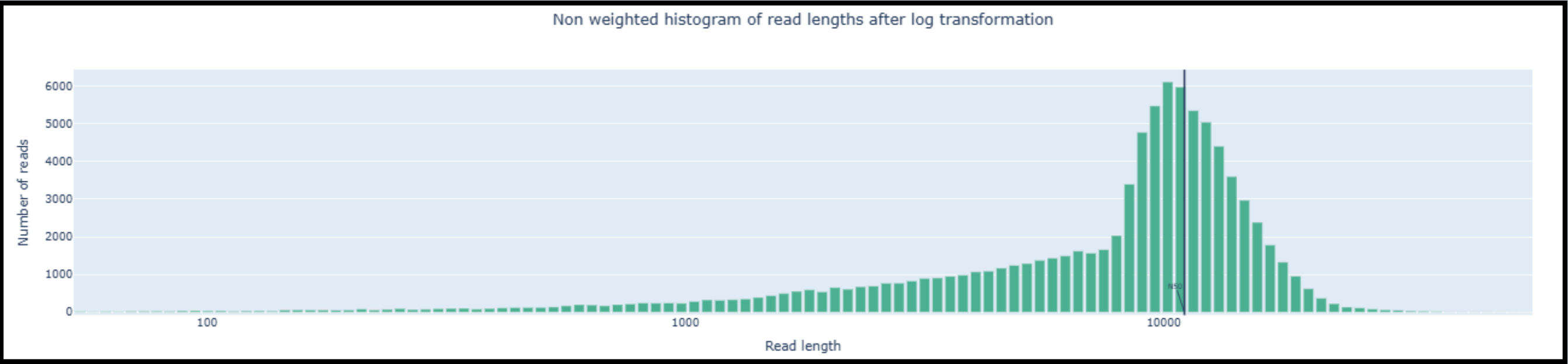
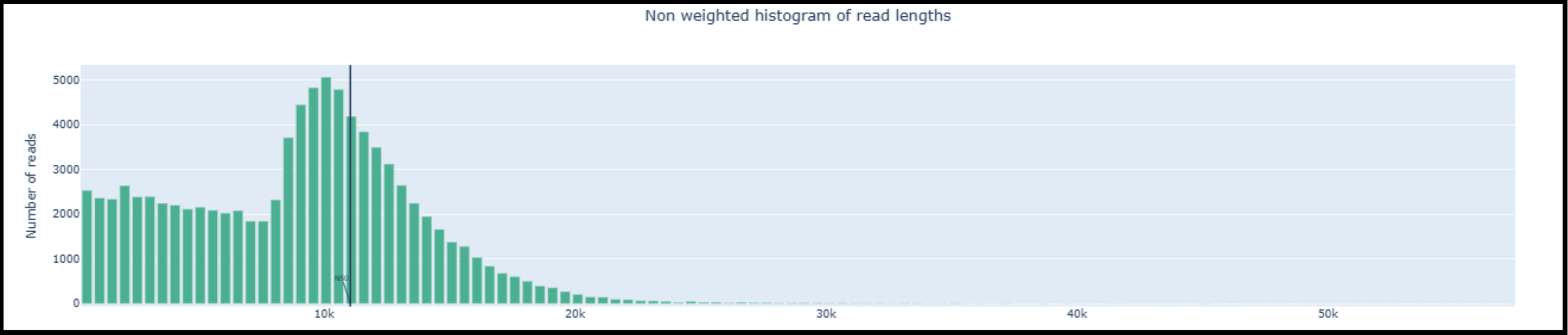
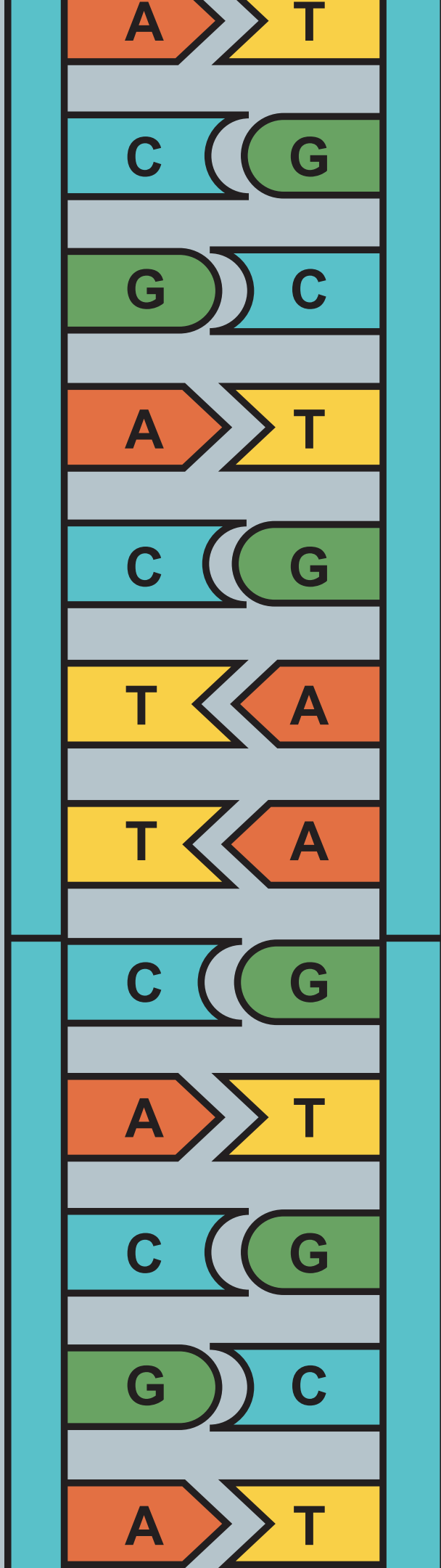
Metrics	
number_of_reads	90422
number_of_bases	790664855.0
median_read_length	9228.0
mean_read_length	8744.2
read_length_stdev	4888.6
n50	11047.0
mean_qual	30.0
median_qual	30.0
longest_read_(with_Q):1	57210 (30.0)
longest_read_(with_Q):2	51837 (30.0)
longest_read_(with_Q):3	50953 (30.0)
longest_read_(with_Q):4	50138 (30.0)
longest_read_(with_Q):5	49568 (30.0)
highest_Q_read_(with_length):1	30.0 (275)
highest_Q_read_(with_length):2	30.0 (8315)
highest_Q_read_(with_length):3	30.0 (13822)
highest_Q_read_(with_length):4	30.0 (3982)
highest_Q_read_(with_length):5	30.0 (2708)
Reads >Q10:	90422 (100.0%) 790.7Mb
Reads >Q15:	90422 (100.0%) 790.7Mb
Reads >Q20:	90422 (100.0%) 790.7Mb
Reads >Q25:	90422 (100.0%) 790.7Mb
Reads >Q30:	0 (0.0%) 0.0Mb

Weighted histogram of read lengths



Weighted histogram of read lengths after log transformation





3. Perform the assemblies (one for each different types of files) and evaluate the quality of each using the reference genome

The assemblers used were:

- **FLYE**, for *SRR21079301.fastq*.
- **SPAdes**, *SRR3137854-forward.fastqsanger* and *SRR31378544-reverse.fastqsanger*

 **SPAdes** genome assembler for genomes of regular and single-cell projects (Galaxy Version 4.1.0+galaxy0) ☆    [Run Tool](#)

Tool Parameters

Operation mode *

Assembly and error correction 

To run read error correction, reads should be in FASTQ format.

Single-end or paired-end short-reads

Paired-end: interlaced reads 

It assumes that all samples belong to the same library. If you want to use samples from two different libraries, include the second library as additional set of short-reads.

FASTA/FASTQ file(s): interlaced *

   2: SRR3137854-forward.fastqsanger x 1: SRR31378544-reverse.fastqsanger x 

accepted formats  switch to column select 

 **Flye** de novo assembler for single molecule sequencing reads (Galaxy Version 2.9.5+galaxy1) ☆    [Run Tool](#)

Tool Parameters

Input reads *

   3: SRR21079301.fastq x 

accepted formats  switch to column select 

Mode

Nanopore raw (--nano-raw) 

Number of polishing iterations *

1

Polishing is performed as the final assembly stage. By default, Flye runs one polishing iteration. Additional iterations might correct a small number of extra errors (due to improvements on how reads may align to the corrected assembly). If the parameter is set to 0, the polishing is not performed (--iterations)

FLYE assembly information:

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8
#seq_name	length	cov.	circ.	repeat	mult.	alt_group	g
contig_1	5325387	131	Y	N	1	*	
contig_2	224163	96	Y	N	1	*	

SPAdes result:

AGAGGCAGAAAACGTTGCTTAACGTGAGTTTTTCGTTCCACTGAGCGTCAGACCCCAATAC
GTGCGCAACAACCGTCTTCCGGAGCCTGTCATACGCGTAAACAGCCAGCGCTGGCGCGA
TTTAGCCCCGACGTATCCCCACTGTTGTCATTTCCGCGCAGACGATGACGTCACTGCC
CGGCTGTATGCGCGAGGTTACCGACTGCGGCCTGAGTTTTTTAAATGGCGGAAAATCGTG
TTGAGGCCAACGCCCATAATGCGGGCGGTTGCCCGGCATCCAACGCCATTTCATGGCCATA
TCAATGATTTTCTGGTGCGTACCGGGTTGAGAAGCGGTGTAAGTGAAGTGCAGTTGCCAT
GTTTTACGGCAGTGAGAGCAGAGATAGCGCTGATGTCCGGCGGTGCTTTTGCCGTTACGC
ACCACCCCGTCAGTAGCTGAACAGGAGGGACAGCTGATAGAAACAGAAGCCACTGGAGCA
CCTCAAAAACACCATCATACACTAAATCAGTAAGTTGGCAGCATCACCGGAAC TTCAAAA
CTGCGCAAAAACACGCAAATACGCACAAGAAAGTAGGGCTGCGCTTGGAATTCGGAAAATA
AACCACATCAAAACTACTTTTTAGCCAAATAACGTTTGGAATCACCAGAATGGTGGGAC
AACAGCGGTTTTAGTGCCCTAAATCGTACGTTTTTCATCCAGTTGCCCTCAAACCCCATG
TTCAAGTCAGAATAGTGGACAGGCGGCCAAGAACTTCGTTTCATGATAGTCTCCGGAACCC
GTTTCGAGTCGTTTTCCGCCCCGTGCTTTCATATCAATTGTCCGGGGTTGATCGCAACGTA
CAACACCTGTGGTACGTATGCCAACACCATCCAACGACACCGCAAAGCCGGCAGTGCGGG
CAAAATTGCCTCCGCTGGTTACGGGCACAACAACAGGCAGGCGGGTCACGCGATTAAAGG
CCGCCGGTGTGACAATCAGCACCGGCCGCGTTCCCTGCTGCTCATGACCTGCGGTAGGAT
CAAGCGAGACAAGCCAGATTTCCCCTCTTTCATGTGAGATTTCTCCTGACCAGTCGCC
GGTGCATCCAGCCATTCTCGTTCTTCAGCTGATATTTAGCATTCGGATCACACTGTGCC
AGTAGCTCAGCCAGTGAATATTGCGGGCGTCTGTACGGCTCAACAATCAGCCGGCCATTA
TCAATGATTTTCTGGTGCGTACCGGGTTGAGAAGCGGTGTAAGTGAAGTGCAGTTGCCAT

3.65 GB

2524

a 46: Circos plot

48: Quast on data 37 and dat

a 46: HTML report

47: SPAdes on data 2 and dat

a 1: Scaffolds

46: SPAdes on data 2 and dat

a 1: Contigs

45: SPAdes on data 2 and dat

a 1: Assembly graph with scaf
folds

Quast

Genome assembly Quality (Galaxy Version 5.3.0+galaxy0)

★

▼

▶ Run Tool

Tool Parameters

Assembly mode?

Individual assembly (1 contig file per sample) ▼

Useful to know if contigs have been generated all samples together (co-assembly) or on each sample individually (individual assembly)

Use customized names for the input files?

No, use dataset names ▼

They will be used in reports, plots and logs

Contigs/scaffolds file *

...

25: Flye on data 3: consensus

▼

accepted formats ▼

Quast

Genome assembly Quality (Galaxy Version 5.3.0+galaxy0)

★

▼

▶ Run Tool

Tool Parameters

Assembly mode?

Individual assembly (1 contig file per sample) ▼

Useful to know if contigs have been generated all samples together (co-assembly) or on each sample individually (individual assembly)

Use customized names for the input files?

No, use dataset names ▼

They will be used in reports, plots and logs

Contigs/scaffolds file *

...

46: SPAdes on data 2 and data 1: Contigs

▼

accepted formats ▼

Use a reference genome?

Yes

Many metrics can't be evaluated without a reference. If this is not what you want, you can skip this step.

Reference genome *

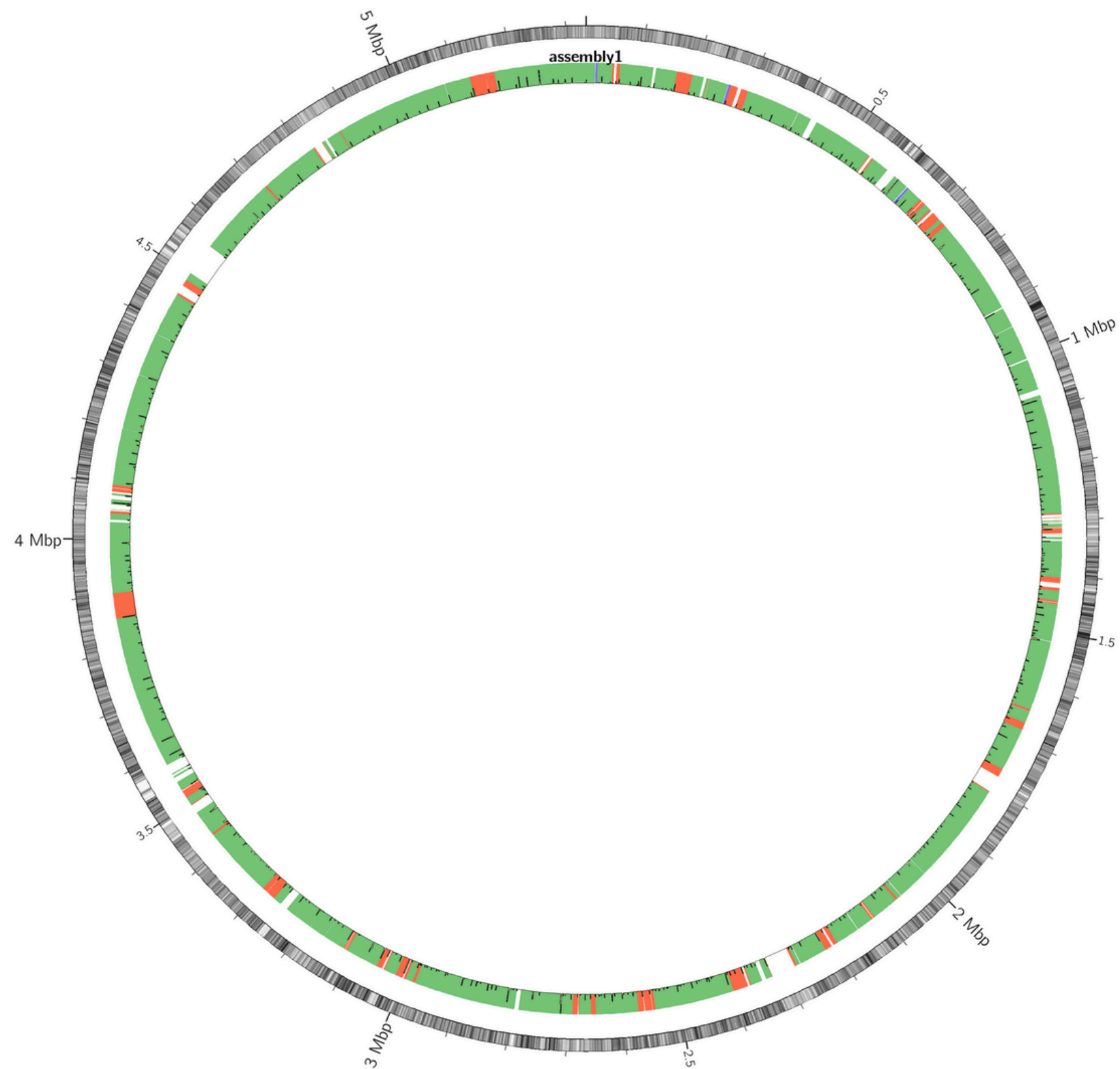
...

37: Kpneumoniae.fasta x

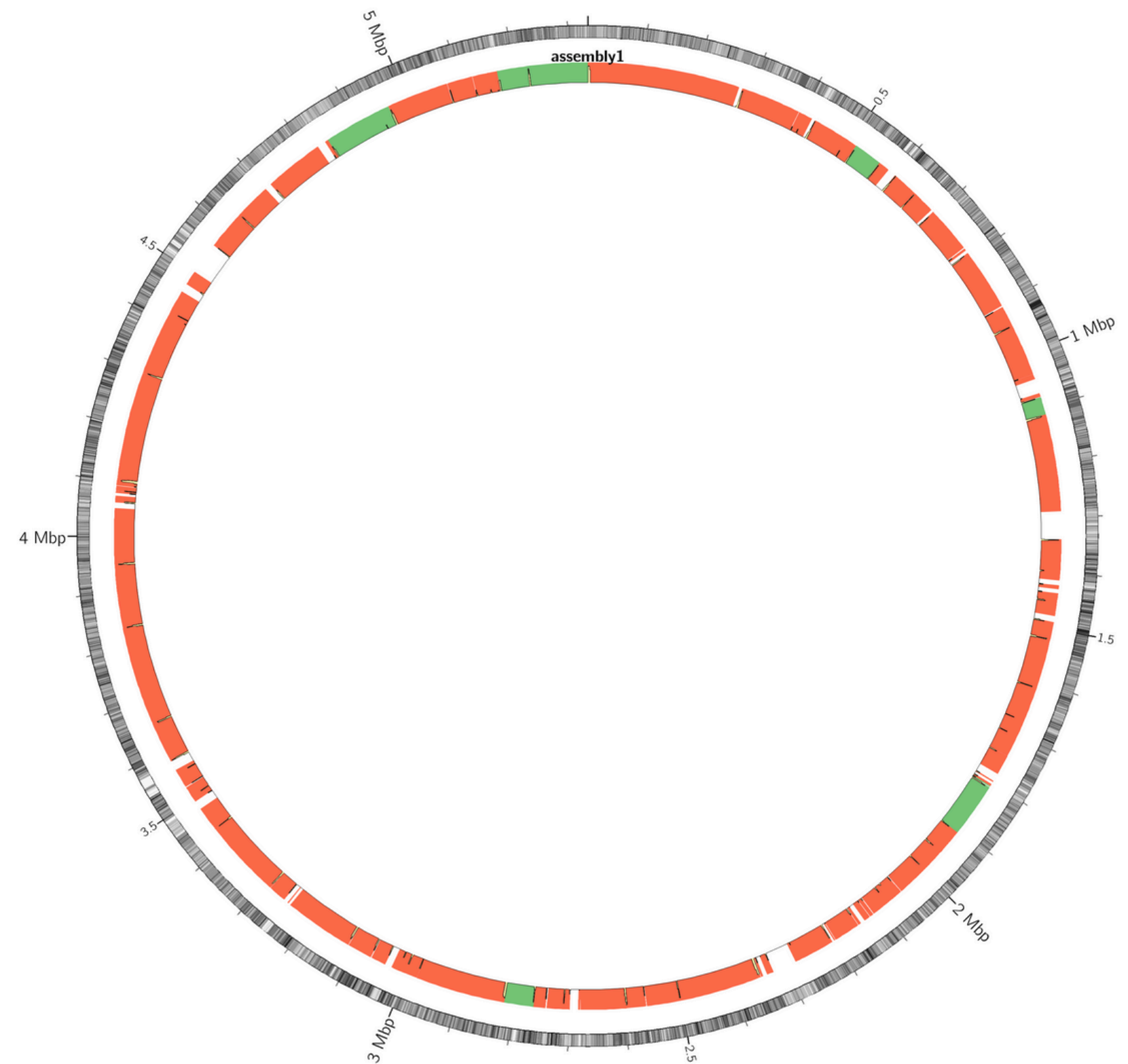
accepted formats ▼

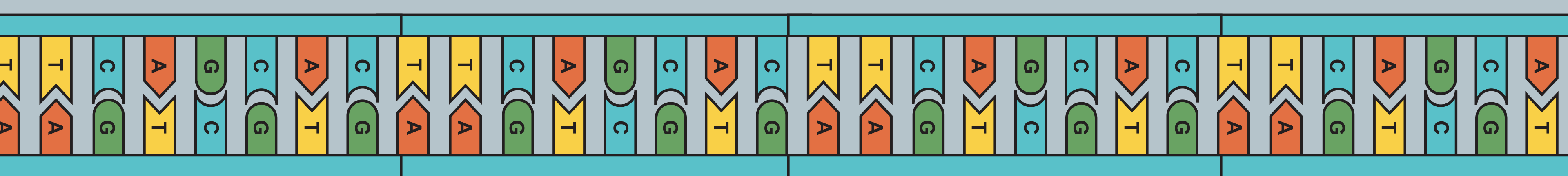
switch to column select ▼

IN SPADES



IN FLYE





Alignment-based statistics		SPAdes_on_data_2_and_data_1_...
Genome fraction (%)		90.923
Duplication ratio		1.002
Largest alignment		47 273
Total aligned length		4 843 067
NGA50		11 823
LGA50		130
Misassemblies		
# misassemblies		35
Misassembled contigs length		558 511
Per base quality		
# mismatches per 100 kbp		660.88
# indels per 100 kbp		16.48
# N's per 100 kbp		0
Statistics without reference		
# contigs		742
Largest contig		58 307
Total length		5 504 720
Total length (>= 1000 bp)		5 437 084
Extended report		

IN SPADES

Alignment-based statistics		Flye_on_data_3__consensus
Genome fraction (%)		91.338
Duplication ratio		1.01
Largest alignment		442 966
Total aligned length		4 864 535
NGA50		104 487
LGA50		16
Misassemblies		
# misassemblies		84
Misassembled contigs length		5 325 387
Per base quality		
# mismatches per 100 kbp		674.76
# indels per 100 kbp		31.7
# N's per 100 kbp		0
Statistics without reference		
# contigs		2
Largest contig		5 325 387
Total length		5 549 550
Total length (>= 1000 bp)		5 549 550
Extended report		

IN FLYE

4. Justify which one would be the best assembly

According to QUAST, an analyzer of the genome assembly quality, the best assembly would have been the one which was obtained from Illumina and done in SPAdes. We can clearly see it in the circos plot from one look. This can be due to several factors:

- For starters, Nanopore has a higher error rate than Illumina, making it easier for an incorrect assembly. However as we saw, the quality of the reads was good, so this should not be the problem.
- Another thing to take into account is the fact that FLAYE has a tendency to produce assemblies which are more complete, but with a higher chance of making incorrect rearrangements, while SPAdes tries to produce more fragmented but accurate assemblies.
- As we can see in the QUAST analysis, the number of missassemblies is considerably higher in the FLAYE than in the SPAdes, indicating that the genome has been assembled incorrectly or that the genome is fragmented.
- Another thing that indicates us that the assembly has not been the best is the high number of mismatches and indels in FLAYE, telling us that there have been substantial errors in the assembly that need to be polished for a better result in this assembler.